

KALASIN STARCH PLANT - JET COOKER PROJECT

PLC SYSTEM CONTROL CABINET CA1 & I/O WIRING SCHEMATICS

Rockwell Automation ControlLogix 1756 Architecture | 1:1 Metric Standard

PROJECT SPECIFICATION & STAKEHOLDERS

Client / Owner:	INGREDION (THAILAND) CO., LTD.
Plant Location:	Kalasin Starch Plant, Thailand
Project Name:	Jet Cooker Automation & Control System
Project Code:	R5THS00138-JC / xPrj-2603001
System Integrator:	AEC INDUSTRIAL ENGINEERING CO., LTD.
Lead Controller:	Allen-Bradley ControlLogix 1756-L950TPSXT
I/O Architecture:	5x 1756 Chassis (C1..C5) + Remote I/O
Main Panel Designation:	CA1 (Main Control Cabinet) / 4-Bay Suite
Total I/O Points:	1,211 Documented Channels
Drawing Standard:	AutoCAD AC1015 / ISO 3098 Metric Standard
Document Status:	APPROVED FOR CONSTRUCTION (Rev 3.6)

DRAWING PACKAGE REVISION RECORD

REV	DATE	DESCRIPTION	BY	APPR
Rev 1.0	2026-06-15	Initial Engineering Concept Draft	AEC	ENG
Rev 2.0	2026-07-10	Chassis & Slot Allocation Baseline	AEC	LEAD
Rev 3.0	2026-08-01	Updated with Client Tagging Dev35	AEC	INGR
Rev 3.5	2026-08-25	Aligned with P&ID and JB Destinations	AEC	PE
Rev 3.6	2026-09-08	Approved I/O List -6 Master Schematics	AEC	MGR

PREPARED BY

AEC INDUSTRIAL ENG.

Instrumentation Dept.

CHECKED BY

LEAD AUTOMATION ENG.

System Architecture

VERIFIED BY

PROJECT ELECTRICAL PE

Licensed Professional Eng.

APPROVED BY

INGREDION PROJECT MGR.

Client Automation Lead

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECT

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

COVER SHEET & PROJECT DIRECTORY
MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756

DWG NO: KAL-JC-WIR-00001
REV: 3.6 DATE: 2026-09-08

DRAWN: AEC-ENG
CHECKED: LEAD-PE

STATUS: APPROVED
SCALE: N.T.S.

PROJECT NO:
R5THS00138-JC

SHEET NO:
01 / 40

DOCUMENT INDEX - COMPLETE DRAWING LIST (KALASIN JET COOKER)

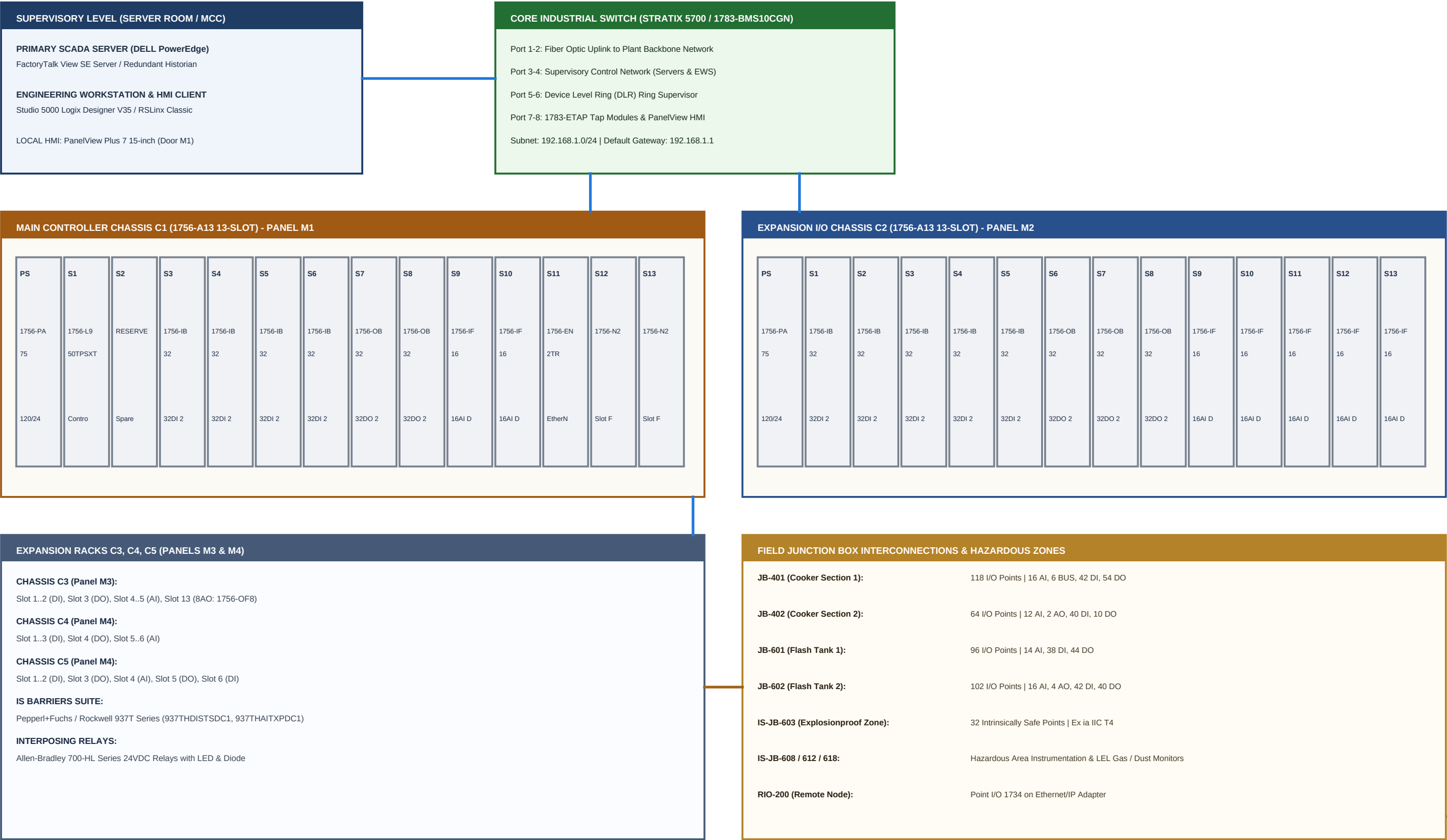
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01	KAL-JC-WIR-00001	COVER SHEET & PROJECT DIRECTORY	3.6	APPROVED
02	KAL-JC-WIR-00002	DRAWING INDEX & SHEET DIRECTORY	3.6	APPROVED
03	KAL-JC-SCD-10001	SYSTEM CONFIGURATION DIAGRAM (SCD)	3.6	APPROVED
04	KAL-JC-WIR-20001	CHASSIS C1 SLOT 3 - 1756-IB32 (DI)	3.6	APPROVED
05	KAL-JC-WIR-20002	CHASSIS C1 SLOT 4 - 1756-IB32 (DI)	3.6	APPROVED
06	KAL-JC-WIR-20003	CHASSIS C1 SLOT 5 - 1756-IB32 (DI)	3.6	APPROVED
07	KAL-JC-WIR-20004	CHASSIS C1 SLOT 6 - 1756-IB32 (DI)	3.6	APPROVED
08	KAL-JC-WIR-20005	CHASSIS C1 SLOT 7 - 1756-OB32 (DO)	3.6	APPROVED
09	KAL-JC-WIR-20006	CHASSIS C1 SLOT 8 - 1756-OB32 (DO)	3.6	APPROVED
10	KAL-JC-WIR-20007	CHASSIS C1 SLOT 9 - 1756-IF16 (AI)	3.6	APPROVED
11	KAL-JC-WIR-20008	CHASSIS C1 SLOT 10 - 1756-IF16 (AI)	3.6	APPROVED
12	KAL-JC-WIR-20009	CHASSIS C2 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
13	KAL-JC-WIR-20010	CHASSIS C2 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
14	KAL-JC-WIR-20011	CHASSIS C2 SLOT 3 - 1756-IB32 (DI)	3.6	APPROVED
15	KAL-JC-WIR-20012	CHASSIS C2 SLOT 4 - 1756-IB32 (DI)	3.6	APPROVED
16	KAL-JC-WIR-20013	CHASSIS C2 SLOT 5 - 1756-IB32 (DI)	3.6	APPROVED
17	KAL-JC-WIR-20014	CHASSIS C2 SLOT 6 - 1756-OB32 (DO)	3.6	APPROVED
18	KAL-JC-WIR-20015	CHASSIS C2 SLOT 7 - 1756-OB32 (DO)	3.6	APPROVED
19	KAL-JC-WIR-20016	CHASSIS C2 SLOT 8 - 1756-OB32 (DO)	3.6	APPROVED
20	KAL-JC-WIR-20017	CHASSIS C2 SLOT 9 - 1756-IF16 (AI)	3.6	APPROVED
21	KAL-JC-WIR-20018	CHASSIS C2 SLOT 10 - 1756-IF16 (AI)	3.6	APPROVED

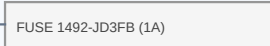
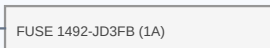
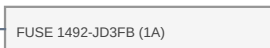
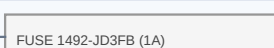
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22	KAL-JC-WIR-20019	CHASSIS C2 SLOT 11 - 1756-IF16 (AI)	3.6	APPROVED
23	KAL-JC-WIR-20020	CHASSIS C2 SLOT 12 - 1756-IF16 (AI)	3.6	APPROVED
24	KAL-JC-WIR-20021	CHASSIS C2 SLOT 13 - 1756-IF16 (AI)	3.6	APPROVED
25	KAL-JC-WIR-20022	CHASSIS C3 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
26	KAL-JC-WIR-20023	CHASSIS C3 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
27	KAL-JC-WIR-20024	CHASSIS C3 SLOT 3 - 1756-OB32 (DO)	3.6	APPROVED
28	KAL-JC-WIR-20025	CHASSIS C3 SLOT 4 - 1756-IF16 (AI)	3.6	APPROVED
29	KAL-JC-WIR-20026	CHASSIS C3 SLOT 5 - 1756-IF16 (AI)	3.6	APPROVED
30	KAL-JC-WIR-20027	CHASSIS C3 SLOT 13 - 1756-OF8 (AO)	3.6	APPROVED
31	KAL-JC-WIR-20028	CHASSIS C4 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
32	KAL-JC-WIR-20029	CHASSIS C4 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
33	KAL-JC-WIR-20030	CHASSIS C4 SLOT 3 - 1756-IB32 (DI)	3.6	APPROVED
34	KAL-JC-WIR-20031	CHASSIS C4 SLOT 4 - 1756-OB32 (DO)	3.6	APPROVED
35	KAL-JC-WIR-20032	CHASSIS C4 SLOT 5 - 1756-IF16 (AI)	3.6	APPROVED
36	KAL-JC-WIR-20033	CHASSIS C4 SLOT 6 - 1756-IF16 (AI)	3.6	APPROVED
37	KAL-JC-WIR-20034	CHASSIS C5 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
38	KAL-JC-WIR-20035	CHASSIS C5 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
39	KAL-JC-WIR-20036	CHASSIS C5 SLOT 3 - 1756-OB32 (DO)	3.6	APPROVED
40	KAL-JC-WIR-20037	CHASSIS C5 SLOT 4 - 1756-IF16 (AI)	3.6	APPROVED

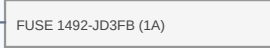
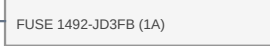
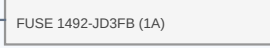
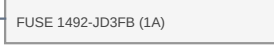
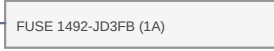
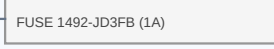
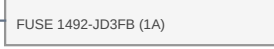
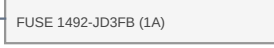
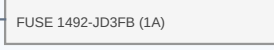
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
DRAWING INDEX & SHEET DIRECTORY MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-00002 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 02 / 40

KALASIN JET COOKER AUTOMATION ARCHITECTURE - CONTROLLOGIX DLR & RIO

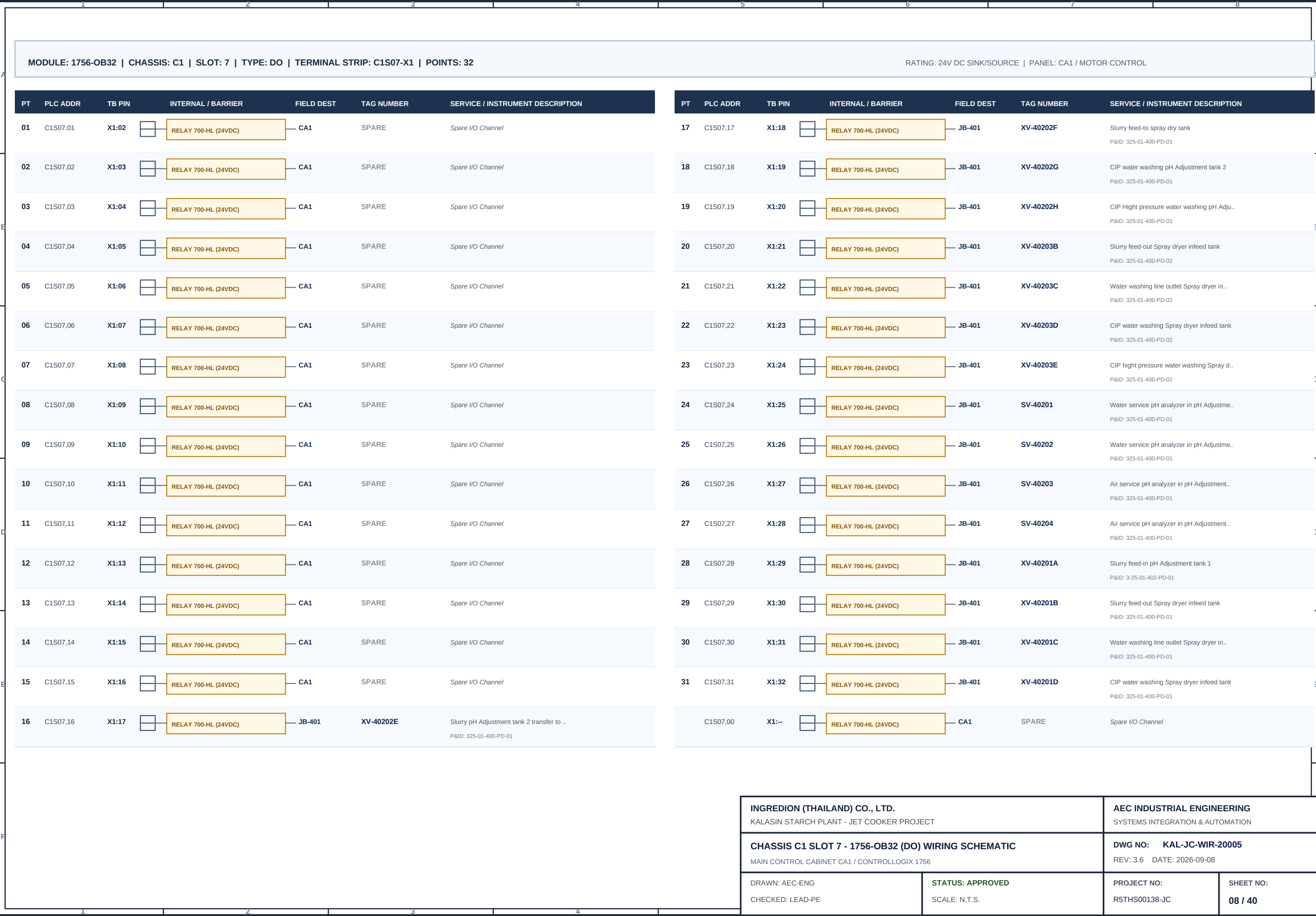
High-Availability Redundant Fiber Optic Device Level Ring (DLR) & Stratix Switch Network



MODULE: 1756-IB32 CHASSIS: C1 SLOT: 3 TYPE: DI TERMINAL STRIP: C1S03-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL																			
PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION				PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01								C1S03,01	X1:02			CA1	SPARE	Spare I/O Channel				17								C1S03,17	X1:18			JB-401	FSL-40202	Slurry low flow P&ID: 3-25-01-402-PD-01			
02								C1S03,02	X1:03			CA1	SPARE	Spare I/O Channel				18								C1S03,18	X1:19			JB-401	FT-40201	Slurry flowrate & density check transfer.. P&ID: 325-01-400-PD-02			
03								C1S03,03	X1:04			CA1	SPARE	Spare I/O Channel				19								C1S03,19	X1:20			JB-401	FT-40202	Feed Slurry Flow Transmitter for Pipe (P.. P&ID: 325-01-400-PD-02			
04								C1S03,04	X1:05			CA1	SPARE	Spare I/O Channel				20								C1S03,20	X1:21			JB-401	HV-40201	Slurry Drain line P&ID: 325-01-400-PD-01			
05								C1S03,05	X1:06			CA1	SPARE	Spare I/O Channel				21								C1S03,21	X1:22			JB-401	HV-40202	Slurry Drain line P&ID: 325-01-400-PD-01			
06								C1S03,06	X1:07			CA1	SPARE	Spare I/O Channel				22								C1S03,22	X1:23			JB-401	HV-40203	Slurry Drain tank TS-40201 P&ID: 325-01-400-PD-01			
07								C1S03,07	X1:08			CA1	SPARE	Spare I/O Channel				23								C1S03,23	X1:24			JB-401	HV-40204	Slurry Drain tank TS-40202 P&ID: 325-01-400-PD-01			
08								C1S03,08	X1:09			CA1	SPARE	Spare I/O Channel				24								C1S03,24	X1:25			JB-401	HV-40205	Slurry Drain line P&ID: 325-01-400-PD-01			
09								C1S03,09	X1:10			CA1	SPARE	Spare I/O Channel				25								C1S03,25	X1:26			JB-401	HV-40206	Slurry Drain tank TS-40203 P&ID: 325-01-400-PD-01			
10								C1S03,10	X1:11			CA1	SPARE	Spare I/O Channel				26								C1S03,26	X1:27			JB-401	XV-40201A	Slurry feed-in pH Adjustment tank 1 P&ID: 3-25-01-402-PD-01			
11								C1S03,11	X1:12			CA1	SPARE	Spare I/O Channel				27								C1S03,27	X1:28			JB-401	XV-40201A	Slurry feed-in pH Adjustment tank 1 P&ID: 3-25-01-402-PD-01			
12								C1S03,12	X1:13			CA1	SPARE	Spare I/O Channel				28								C1S03,28	X1:29			JB-401	XV-40201B	Slurry feed-out Spray dryer infeed tank P&ID: 325-01-400-PD-01			
13								C1S03,13	X1:14			CA1	SPARE	Spare I/O Channel				29								C1S03,29	X1:30			JB-401	XV-40201B	Slurry feed-out Spray dryer infeed tank P&ID: 325-01-400-PD-01			
14								C1S03,14	X1:15			CA1	SPARE	Spare I/O Channel				30								C1S03,30	X1:31			JB-401	XV-40201C	Water washing line outlet Spray dryer in.. P&ID: 325-01-400-PD-01			
15								C1S03,15	X1:16			CA1	SPARE	Spare I/O Channel				31								C1S03,31	X1:32			JB-401	XV-40201C	Water washing line outlet Spray dryer in.. P&ID: 325-01-400-PD-01			
16								C1S03,16	X1:17			JB-401	FSL-40201	Slurry low flow P&ID: 3-25-01-402-PD-01				C1S03,00								X1:--			CA1	SPARE	Spare I/O Channel				
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT																AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION																			
CHASSIS C1 SLOT 3 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756																DWG NO: KAL-JC-WIR-20001 REV: 3.6 DATE: 2026-09-08																			
DRAWN: AEC-ENG CHECKED: LEAD-PE								STATUS: APPROVED SCALE: N.T.S.								PROJECT NO: R5THS00138-JC				SHEET NO: 04 / 40															

MODULE: 1756-IB32 CHASSIS: C1 SLOT: 5 TYPE: DI TERMINAL STRIP: C1S05-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL															
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION															
01	C1S05,01	X1:02		 FUSE 1492-JD3FB (1A)	JB-401	XV-40203E	CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-02		17	C1S05,17	X1:18		 FUSE 1492-JD3FB (1A)	JB-601	FS-60205	Cooling Water Return Flow Switch for Pos.. P&ID: 325-01-600-PD-01															
02	C1S05,02	X1:03		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		18	C1S05,18	X1:19		 FUSE 1492-JD3FB (1A)	JB-601	LSL-60201	Level low switch P&ID: 325-01-600-PD-11															
03	C1S05,03	X1:04		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		19	C1S05,19	X1:20		 FUSE 1492-JD3FB (1A)	JB-601	SV-60201	PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08															
04	C1S05,04	X1:05		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		20	C1S05,20	X1:21		 FUSE 1492-JD3FB (1A)	JB-601	SV-60201	PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08															
05	C1S05,05	X1:06		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		21	C1S05,21	X1:22		 FUSE 1492-JD3FB (1A)	JB-601	SV-60202	PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08															
06	C1S05,06	X1:07		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		22	C1S05,22	X1:23		 FUSE 1492-JD3FB (1A)	JB-601	SV-60202	PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08															
07	C1S05,07	X1:08		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		23	C1S05,23	X1:24		 FUSE 1492-JD3FB (1A)	JB-601	SV-60203	PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08															
08	C1S05,08	X1:09		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		24	C1S05,24	X1:25		 FUSE 1492-JD3FB (1A)	JB-601	SV-60203	PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08															
09	C1S05,09	X1:10		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		25	C1S05,25	X1:26		 FUSE 1492-JD3FB (1A)	JB-601	VS-60205	Vibration Gear Box Vibration Switch for .. P&ID: 325-01-600-PD-01															
10	C1S05,10	X1:11		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		26	C1S05,26	X1:27		 FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel															
11	C1S05,11	X1:12		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		27	C1S05,27	X1:28		 FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel															
12	C1S05,12	X1:13		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		28	C1S05,28	X1:29		 FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel															
13	C1S05,13	X1:14		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		29	C1S05,29	X1:30		 FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel															
14	C1S05,14	X1:15		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		30	C1S05,30	X1:31		 FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel															
15	C1S05,15	X1:16		 FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		31	C1S05,31	X1:32		 FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel															
16	C1S05,16	X1:17		 FUSE 1492-JD3FB (1A)	JB-601	ES-60202	E-Stop for PUPD-11 (Positive Displaceme.. P&ID: 325-01-600-PD-01			C1S05,00	X1:--		 FUSE 1492-JD3FB (1A)	JB-401	XV-40203E	CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-02															

MODULE: 1756-IB32 CHASSIS: C1 SLOT: 6 TYPE: DI TERMINAL STRIP: C1S06-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL															
PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01							C1S06,01	X1:02	FUSE 1492-JD3FB (1A)	JB-601	XA-60202B	General Fault Controlled Stop for XSUP-5.. P&ID: 325-01-600-PD-08		17							C1S06,17	X1:18	FUSE 1492-JD3FB (1A)	JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-05					
02							C1S06,02	X1:03	FUSE 1492-JD3FB (1A)	JB-601	XA-60202C	General Armed Status Indicator for XSUP-.. P&ID: 325-01-600-PD-08		18							C1S06,18	X1:19	FUSE 1492-JD3FB (1A)	JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-08					
03							C1S06,03	X1:04	FUSE 1492-JD3FB (1A)	JB-601	XA-60202D	Power Fault Status Indicator for XSUP-56.. P&ID: 325-01-600-PD-08		19							C1S06,19	X1:20	FUSE 1492-JD3FB (1A)	JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-05					
04							C1S06,04	X1:05	FUSE 1492-JD3FB (1A)	JB-601	XA-60202E	Dryer CHAM-21 Emergency Stop Actuation f.. P&ID: 325-01-600-PD-08		20							C1S06,20	X1:21	FUSE 1492-JD3FB (1A)	JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-08					
05							C1S06,05	X1:06	FUSE 1492-JD3FB (1A)	JB-601	XA-60202F	CHAM-21 Controlled Stop Trouble (No Prot.. P&ID: 325-01-600-PD-08		21							C1S06,21	X1:22	FUSE 1492-JD3FB (1A)	JB-601	YA-60205	Fault Command P&ID: 325-01-600-PD-01					
06							C1S06,06	X1:07	FUSE 1492-JD3FB (1A)	JB-601	XA-60202G	CYCL-21/22 Emergency Stop Actuation for .. P&ID: 325-01-600-PD-08		22							C1S06,22	X1:23	FUSE 1492-JD3FB (1A)	JB-601	YI-60205	Run Command P&ID: 325-01-600-PD-01					
07							C1S06,07	X1:08	FUSE 1492-JD3FB (1A)	JB-601	XA-60202H	CYCL-21/22 Controlled Stop Trouble (No P.. P&ID: 325-01-600-PD-08		23							C1S06,23	X1:24	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
08							C1S06,08	X1:09	FUSE 1492-JD3FB (1A)	JB-601	XA-60202I	Instrument XA-60202I P&ID: 325-01-600-PD-08		24							C1S06,24	X1:25	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
09							C1S06,09	X1:10	FUSE 1492-JD3FB (1A)	JB-601	XA-60202J	BGHS-21 Emergency Stop Actuation for XSU.. P&ID: 325-01-600-PD-08		25							C1S06,25	X1:26	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
10							C1S06,10	X1:11	FUSE 1492-JD3FB (1A)	JB-601	XA-60202K	BGHS-21 Controlled Stop Trouble (No Prot.. P&ID: 325-01-600-PD-08		26							C1S06,26	X1:27	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
11							C1S06,11	X1:12	FUSE 1492-JD3FB (1A)	JB-601	XA-60202L	CYCL-31 Emergency Stop Actuation for XSU.. P&ID: 325-01-600-PD-08		27							C1S06,27	X1:28	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
12							C1S06,12	X1:13	FUSE 1492-JD3FB (1A)	JB-601	XA-60202M	CYCL-31 Controlled Stop Trouble (No Prot.. P&ID: 325-01-600-PD-08		28							C1S06,28	X1:29	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
13							C1S06,13	X1:14	FUSE 1492-JD3FB (1A)	JB-601	XV-60202	High Pressure Water On/Off Valve for Dra.. P&ID: 325-01-600-PD-11		29							C1S06,29	X1:30	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
14							C1S06,14	X1:15	FUSE 1492-JD3FB (1A)	JB-601	XV-60202	High Pressure Water On/Off Valve for Dra.. P&ID: 325-01-600-PD-11		30							C1S06,30	X1:31	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
15							C1S06,15	X1:16	FUSE 1492-JD3FB (1A)	JB-601	XV-60205A	Cooling Water On/Off Valve for Cooling W.. P&ID: 325-01-600-PD-01		31							C1S06,31	X1:32	FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel					
16							C1S06,16	X1:17	FUSE 1492-JD3FB (1A)	JB-601	XV-60205A	Cooling Water On/Off Valve for Cooling W.. P&ID: 325-01-600-PD-01		C1S06,00							X1:--	FUSE 1492-JD3FB (1A)	JB-601	XA-60202A	General Alarm Emergency Stop Actuation f.. P&ID: 325-01-600-PD-08						
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT																AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION															
CHASSIS C1 SLOT 6 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756																DWG NO: KAL-JC-WIR-20004 REV: 3.6 DATE: 2026-09-08															
DRAWN: AEC-ENG CHECKED: LEAD-PE								STATUS: APPROVED SCALE: N.T.S.								PROJECT NO: R5THS00138-JC				SHEET NO: 07 / 40											



MODULE: 1756-OB32 CHASSIS: C1 SLOT: 8 TYPE: DO TERMINAL STRIP: C1S08-X1 POINTS: 32								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C1S08,01	X1:02		RELAY 700-HL (24VDC)	JB-401	XV-40201F	Slurry feed-to spray dry tank P&ID: 325-01-400-PD-01	17	C1S08,17	X1:18		RELAY 700-HL (24VDC)	JB-601	SV-60202	PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08
02	C1S08,02	X1:03		RELAY 700-HL (24VDC)	JB-401	XV-40201G	CIP water washing pH Adjustment tank 1 P&ID: 325-01-400-PD-01	18	C1S08,18	X1:19		RELAY 700-HL (24VDC)	JB-601	SV-60230	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-04
03	C1S08,03	X1:04		RELAY 700-HL (24VDC)	JB-401	XV-40201H	CIP Hight pressure water washing pH Adju.. P&ID: 325-01-400-PD-01	19	C1S08,19	X1:20		RELAY 700-HL (24VDC)	JB-601	SV-60203	PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08
04	C1S08,04	X1:05		RELAY 700-HL (24VDC)	JB-401	XV-40202A	Slurry feed-in pH Adjustment tank 2 P&ID: 325-01-400-PD-01	20	C1S08,20	X1:21		RELAY 700-HL (24VDC)	JB-601	SV-60240A	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-05
05	C1S08,05	X1:06		RELAY 700-HL (24VDC)	JB-401	XV-40202B	Slurry feed-out pH Adjustment tank 2 P&ID: 325-01-400-PD-01	21	C1S08,21	X1:22		RELAY 700-HL (24VDC)	JB-601	SV-60231	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-04
06	C1S08,06	X1:07		RELAY 700-HL (24VDC)	JB-401	XV-40202C	Water washing line outlet pH Adjustment .. P&ID: 325-01-400-PD-01	22	C1S08,22	X1:23		RELAY 700-HL (24VDC)	JB-601	SV-60240B	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-05
07	C1S08,07	X1:08		RELAY 700-HL (24VDC)	JB-401	XV-40202D	Slurry circulation pH Adjustment tank 2 P&ID: 325-01-400-PD-01	23	C1S08,23	X1:24		RELAY 700-HL (24VDC)	JB-601	SV-60240C	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-05
08	C1S08,08	X1:09		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	24	C1S08,24	X1:25		RELAY 700-HL (24VDC)	JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-05
09	C1S08,09	X1:10		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	25	C1S08,25	X1:26		RELAY 700-HL (24VDC)	JB-601	XV-60251	Liquid On/Off Valve for Drain Pipe (PIPE) P&ID: 325-01-600-PD-01
10	C1S08,10	X1:11		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	26	C1S08,26	X1:27		RELAY 700-HL (24VDC)	JB-601	XV-60202	High Pressure Water On/Off Valve for Dra.. P&ID: 325-01-600-PD-11
11	C1S08,11	X1:12		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	27	C1S08,27	X1:28		RELAY 700-HL (24VDC)	JB-601	XV-60205A	Cooling Water On/Off Valve for Cooling W.. P&ID: 325-01-600-PD-01
12	C1S08,12	X1:13		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	28	C1S08,28	X1:29		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
13	C1S08,13	X1:14		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	29	C1S08,29	X1:30		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
14	C1S08,14	X1:15		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	30	C1S08,30	X1:31		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
15	C1S08,15	X1:16		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	31	C1S08,31	X1:32		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
16	C1S08,16	X1:17		RELAY 700-HL (24VDC)	JB-601	SV-60201	PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08		C1S08,00	X1:--		RELAY 700-HL (24VDC)	JB-401	XV-40201E	CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-01
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT								AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION							
CHASSIS C1 SLOT 8 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756								DWG NO: KAL-JC-WIR-20006 REV: 3.6 DATE: 2026-09-08							
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.				PROJECT NO: R5THS00138-JC		SHEET NO: 09 / 40					

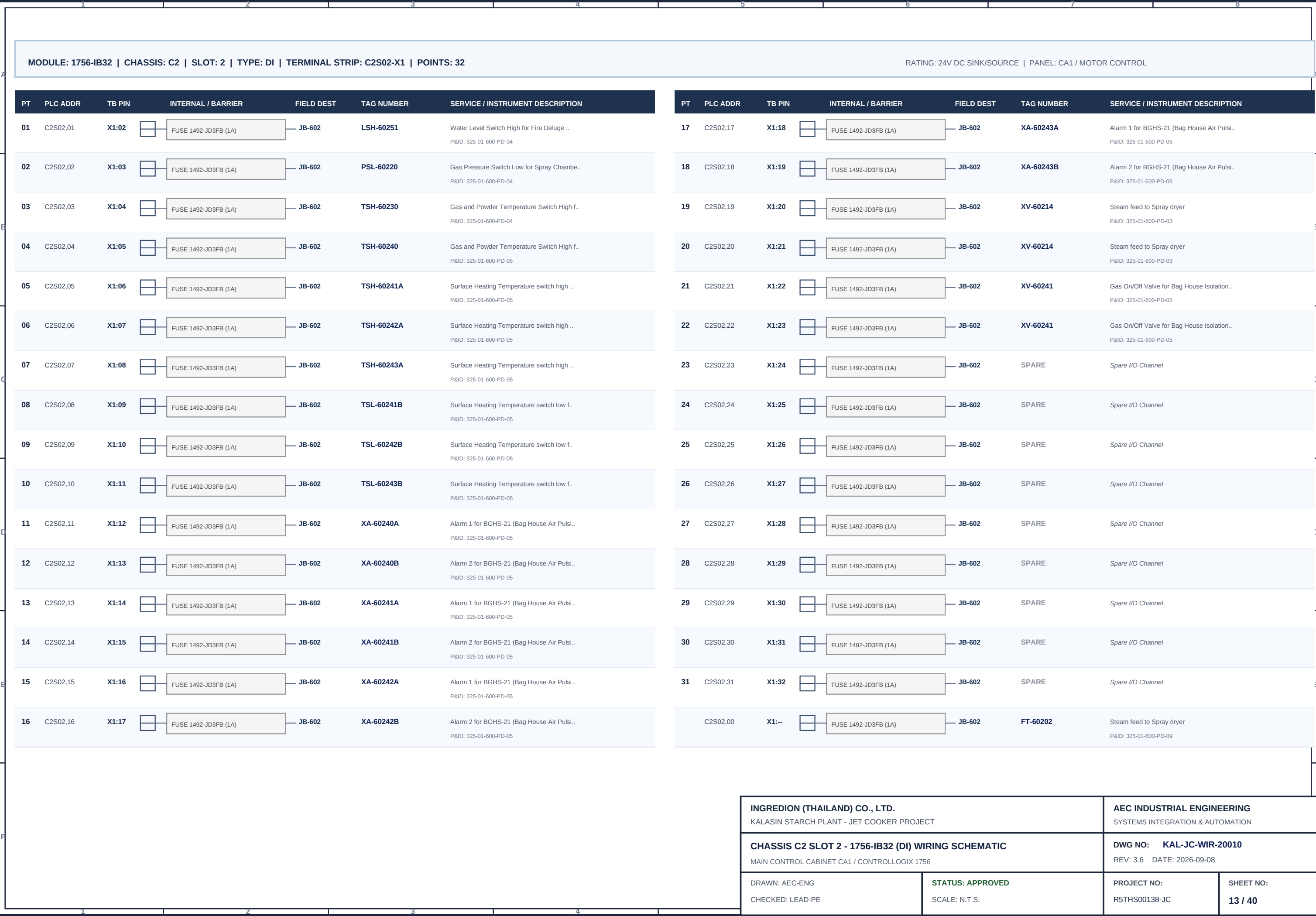
MODULE: 1756-IF16 CHASSIS: C1 SLOT: 9 TYPE: AI TERMINAL STRIP: C1S09-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C1S09,01	X1:02		LOOP POWERED (2-W)	JB-401	FT-40201	Slurry flowrate & density check transfer.. P&ID: 325-01-400-PD-02	
02	C1S09,02	X1:03		LOOP POWERED (2-W)	JB-401	FT-40202	Feed Slurry Flow Transmitter for Pipe (P.. P&ID: 325-01-400-PD-02	
03	C1S09,03	X1:04		LOOP POWERED (2-W)	JB-401	LT-40203	Level transmitter in Spray dryer infeed .. P&ID: 325-01-400-PD-02	
04	C1S09,04	X1:05		LOOP POWERED (2-W)	JB-401	LT-40201	Level transmitter in Spray dryer infeed .. P&ID: 325-01-400-PD-01	
05	C1S09,05	X1:06		LOOP POWERED (2-W)	JB-401	LT-40202	Level transmitter in pH Adjustment tank 2 P&ID: 325-01-400-PD-01	
06	C1S09,06	X1:07		LOOP POWERED (2-W)	JB-401	pHT-40201	pH analyzer in pH Adjustment tank 1 P&ID: 325-01-400-PD-01	
07	C1S09,07	X1:08		LOOP POWERED (2-W)	JB-401	pHT-40202	pH analyzer in pH Adjustment tank 2 P&ID: 325-01-400-PD-01	
08	C1S09,08	X1:09		LOOP POWERED (2-W)	JB-401	PT-40201	Pressure Transmitter for Inlet Filter S.. P&ID: 325-01-400-PD-02	
09	C1S09,09	X1:10		LOOP POWERED (2-W)	JB-401	PT-40202	Feed Slurry Pressure Transmitter for Pip.. P&ID: 325-01-400-PD-02	
10	C1S09,10	X1:11		LOOP POWERED (2-W)	JB-401	PT-60202	Feed Slurry Pressure Transmitter for Hig.. P&ID: 325-01-600-PD-01	
11	C1S09,11	X1:12		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
12	C1S09,12	X1:13		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
13	C1S09,13	X1:14		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
14	C1S09,14	X1:15		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
15	C1S09,15	X1:16		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
	C1S09,00	X1:--		LOOP POWERED (2-W)	JB-401	FT-40201	Slurry flowrate & density check transfer.. P&ID: 325-01-400-PD-02	

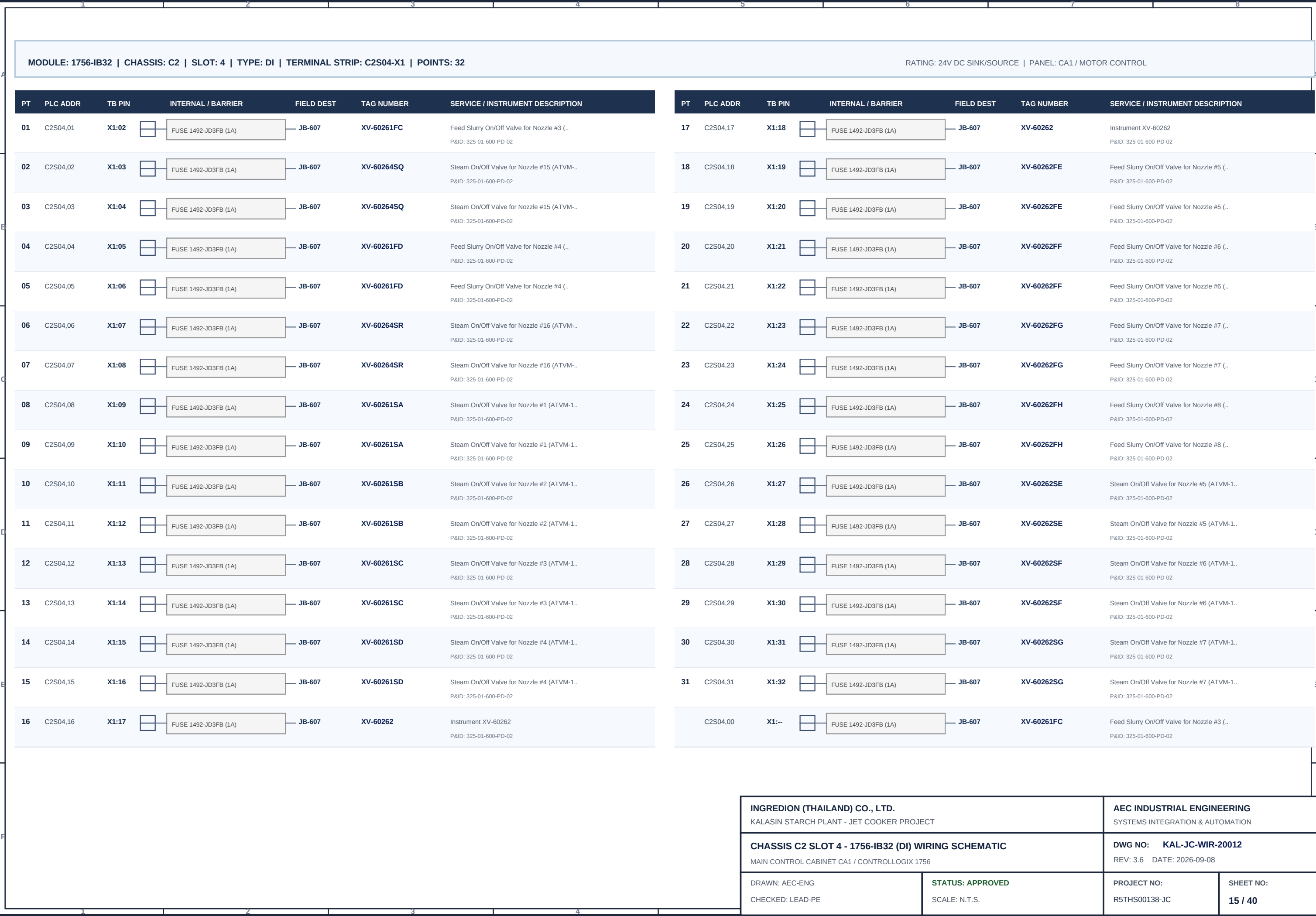
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C1 SLOT 9 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20007 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 10 / 40

MODULE: 1756-IF16 CHASSIS: C1 SLOT: 10 TYPE: AI TERMINAL STRIP: C1S10-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL						
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION						
01	C1S10,01	X1:02		LOOP POWERED (2-W)	JB-601	DPT-60202	Gas Pressure Differential Transmitter f.. P&ID: 325-01-600-PD-08						
02	C1S10,02	X1:03		LOOP POWERED (2-W)	JB-601	FT-60201	Gas Flow Transmitter for Cooling Conveyl.. P&ID: 325-01-600-PD-08						
03	C1S10,03	X1:04		LOOP POWERED (2-W)	JB-601	FT-60204	Compressed air Flow P&ID: 325-01-600-PD-10						
04	C1S10,04	X1:05		LOOP POWERED (2-W)	JB-601	FT-61307	Compressed air Flow P&ID: 3-25-01-613-PD-04						
05	C1S10,05	X1:06		LOOP POWERED (2-W)	JB-601	PT-60203	Gas Pressure Transmitter for Cool Air Du.. P&ID: 325-01-600-PD-08						
06	C1S10,06	X1:07		LOOP POWERED (2-W)	JB-601	PT-60204	Pressure transmitter P&ID: 325-01-600-PD-10						
07	C1S10,07	X1:08		LOOP POWERED (2-W)	JB-601	PT-60205	Feed Slurry Pressure Transmitter for Hig.. P&ID: 325-01-600-PD-01						
08	C1S10,08	X1:09		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
09	C1S10,09	X1:10		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
10	C1S10,10	X1:11		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
11	C1S10,11	X1:12		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
12	C1S10,12	X1:13		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
13	C1S10,13	X1:14		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
14	C1S10,14	X1:15		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
15	C1S10,15	X1:16		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel						
	C1S10,00	X1:--		LOOP POWERED (2-W)	JB-601	DPI-60240	Gas Pressure Differntial Pressure (Bag H.. P&ID: 325-01-600-PD-05						

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C1 SLOT 10 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20008 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 11 / 40

MODULE: 1756-IB32 CHASSIS: C2 SLOT: 1 TYPE: DI TERMINAL STRIP: C2S01-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL																			
PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION				PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01								C2S01,01	X1:02	 FUSE 1492-JD3FB (1A)		JB-402	FS-40205	Jet Cooker P&ID: 325-01-402-PD-02				17								C2S01,17	X1:18	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205C	Jet Cooker P&ID: 325-01-402-PD-02			
02								C2S01,02	X1:03	 FUSE 1492-JD3FB (1A)		JB-402	FT-40221	Jet Cooker P&ID: 325-01-402-PD-02				18								C2S01,18	X1:19	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205D	Jet Cooker P&ID: 325-01-402-PD-02			
03								C2S01,03	X1:04	 FUSE 1492-JD3FB (1A)		JB-402	FT-40222	Jet Cooker P&ID: 325-01-402-PD-02				19								C2S01,19	X1:20	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205D	Jet Cooker P&ID: 325-01-402-PD-02			
04								C2S01,04	X1:05	 FUSE 1492-JD3FB (1A)		JB-402	HV-40207	Jet Cooker P&ID: 325-01-402-PD-01				20								C2S01,20	X1:21	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205F	Jet Cooker P&ID: 325-01-402-PD-02			
05								C2S01,05	X1:06	 FUSE 1492-JD3FB (1A)		JB-402	HV-40208	Jet Cooker P&ID: 325-01-402-PD-01				21								C2S01,21	X1:22	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205F	Jet Cooker P&ID: 325-01-402-PD-02			
06								C2S01,06	X1:07	 FUSE 1492-JD3FB (1A)		JB-402	XV-40203F	Jet Cooker P&ID: 325-01-402-PD-02				22								C2S01,22	X1:23	 FUSE 1492-JD3FB (1A)		JB-402	XV-40222	Jet Cooker P&ID: 325-01-402-PD-02			
07								C2S01,07	X1:08	 FUSE 1492-JD3FB (1A)		JB-402	XV-40203F	Jet Cooker P&ID: 325-01-402-PD-02				23								C2S01,23	X1:24	 FUSE 1492-JD3FB (1A)		JB-402	XV-40222	Jet Cooker			
08								C2S01,08	X1:09	 FUSE 1492-JD3FB (1A)		JB-402	XV-40204B	Jet Cooker P&ID: 325-01-402-PD-01				24								C2S01,24	X1:25	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
09								C2S01,09	X1:10	 FUSE 1492-JD3FB (1A)		JB-402	XV-40204B	Jet Cooker P&ID: 325-01-402-PD-01				25								C2S01,25	X1:26	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
10								C2S01,10	X1:11	 FUSE 1492-JD3FB (1A)		JB-402	XV-40204C	Jet Cooker P&ID: 325-01-402-PD-01				26								C2S01,26	X1:27	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
11								C2S01,11	X1:12	 FUSE 1492-JD3FB (1A)		JB-402	XV-40204C	Jet Cooker P&ID: 325-01-402-PD-01				27								C2S01,27	X1:28	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
12								C2S01,12	X1:13	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205A	Jet Cooker P&ID: 325-01-402-PD-02				28								C2S01,28	X1:29	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
13								C2S01,13	X1:14	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205A	Jet Cooker P&ID: 325-01-402-PD-02				29								C2S01,29	X1:30	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
14								C2S01,14	X1:15	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205B	Jet Cooker P&ID: 325-01-402-PD-02				30								C2S01,30	X1:31	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
15								C2S01,15	X1:16	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205B	Jet Cooker P&ID: 325-01-402-PD-02				31								C2S01,31	X1:32	 FUSE 1492-JD3FB (1A)		JB-402	SPARE	Spare I/O Channel			
16								C2S01,16	X1:17	 FUSE 1492-JD3FB (1A)		JB-402	XV-40205C	Jet Cooker P&ID: 325-01-402-PD-02												C2S01,00	X1:--	 FUSE 1492-JD3FB (1A)		JB-402	FS-40204	Jet Cooker P&ID: 325-01-402-PD-01			
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT																AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION																			
CHASSIS C2 SLOT 1 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756																DWG NO: KAL-JC-WIR-20009 REV: 3.6 DATE: 2026-09-08																			
DRAWN: AEC-ENG CHECKED: LEAD-PE												STATUS: APPROVED SCALE: N.T.S.												PROJECT NO: R5THS00138-JC				SHEET NO: 12 / 40							





MODULE: 1756-IB32 CHASSIS: C2 SLOT: 5 TYPE: DI TERMINAL STRIP: C2S05-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL															
PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01							C2S05,01	X1:02		FUSE 1492-JD3FB (1A)		JB-607	XV-60262SH	Steam On/Off Valve for Nozzle #8 (ATVM-1.. P&ID: 325-01-600-PD-02		17							C2S05,17	X1:18		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SL	Steam On/Off Valve for Nozzle #11 (ATVM-.. P&ID: 325-01-600-PD-02	
02							C2S05,02	X1:03		FUSE 1492-JD3FB (1A)		JB-607	XV-60263	Instrument XV-60263 P&ID: 325-01-600-PD-02		18							C2S05,18	X1:19		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SM	Steam On/Off Valve for Nozzle #12 (ATVM-.. P&ID: 325-01-600-PD-02	
03							C2S05,03	X1:04		FUSE 1492-JD3FB (1A)		JB-607	XV-60263	Instrument XV-60263 P&ID: 325-01-600-PD-02		19							C2S05,19	X1:20		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SM	Steam On/Off Valve for Nozzle #12 (ATVM-.. P&ID: 325-01-600-PD-02	
04							C2S05,04	X1:05		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FJ	Feed Slurry On/Off Valve for Nozzle #9 (.. P&ID: 325-01-600-PD-02		20							C2S05,20	X1:21		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
05							C2S05,05	X1:06		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FJ	Feed Slurry On/Off Valve for Nozzle #9 (.. P&ID: 325-01-600-PD-02		21							C2S05,21	X1:22		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
06							C2S05,06	X1:07		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FK	Feed Slurry On/Off Valve for Nozzle #10 .. P&ID: 325-01-600-PD-02		22							C2S05,22	X1:23		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
07							C2S05,07	X1:08		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FK	Feed Slurry On/Off Valve for Nozzle #10 .. P&ID: 325-01-600-PD-02		23							C2S05,23	X1:24		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
08							C2S05,08	X1:09		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FL	Feed Slurry On/Off Valve for Nozzle #11 .. P&ID: 325-01-600-PD-02		24							C2S05,24	X1:25		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
09							C2S05,09	X1:10		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FL	Feed Slurry On/Off Valve for Nozzle #11 .. P&ID: 325-01-600-PD-02		25							C2S05,25	X1:26		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
10							C2S05,10	X1:11		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FM	Feed Slurry On/Off Valve for Nozzle #12 .. P&ID: 325-01-600-PD-02		26							C2S05,26	X1:27		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
11							C2S05,11	X1:12		FUSE 1492-JD3FB (1A)		JB-607	XV-60263FM	Feed Slurry On/Off Valve for Nozzle #12 .. P&ID: 325-01-600-PD-02		27							C2S05,27	X1:28		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
12							C2S05,12	X1:13		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SJ	Steam On/Off Valve for Nozzle #9 (ATVM-1.. P&ID: 325-01-600-PD-02		28							C2S05,28	X1:29		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
13							C2S05,13	X1:14		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SJ	Steam On/Off Valve for Nozzle #9 (ATVM-1.. P&ID: 325-01-600-PD-02		29							C2S05,29	X1:30		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
14							C2S05,14	X1:15		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SK	Steam On/Off Valve for Nozzle #10 (ATVM-.. P&ID: 325-01-600-PD-02		30							C2S05,30	X1:31		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
15							C2S05,15	X1:16		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SK	Steam On/Off Valve for Nozzle #10 (ATVM-.. P&ID: 325-01-600-PD-02		31							C2S05,31	X1:32		FUSE 1492-JD3FB (1A)		JB-607	SPARE	Spare I/O Channel	
16							C2S05,16	X1:17		FUSE 1492-JD3FB (1A)		JB-607	XV-60263SL	Steam On/Off Valve for Nozzle #11 (ATVM-.. P&ID: 325-01-600-PD-02		C2S05,00							X1:--		FUSE 1492-JD3FB (1A)		JB-607	XV-60262SH	Steam On/Off Valve for Nozzle #8 (ATVM-1.. P&ID: 325-01-600-PD-02		

MODULE: 1756-OB32 CHASSIS: C2 SLOT: 6 TYPE: DO TERMINAL STRIP: C2S06-X1 POINTS: 32														RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL													
DO Channel 1 (Points 01-16)							DO Channel 2 (Points 17-32)																				
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN	INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION														
01	C2S06,01	X1:02	RELAY 700-HL (24VDC)	JB-402	XV-40204B	Jet Cooker P&ID: 325-01-402-PD-01	17	C2S06,17	X1:18	RELAY 700-HL (24VDC)	JB-602	XV-60214	Steam feed to Spray dryer P&ID: 325-01-600-PD-03														
02	C2S06,02	X1:03	RELAY 700-HL (24VDC)	JB-402	XV-40204C	Jet Cooker P&ID: 325-01-402-PD-01	18	C2S06,18	X1:19	RELAY 700-HL (24VDC)	JB-602	XV-60241	Gas On/Off Valve for Bag House Isolation.. P&ID: 325-01-600-PD-05														
03	C2S06,03	X1:04	RELAY 700-HL (24VDC)	JB-402	XV-40205A	Jet Cooker P&ID: 325-01-402-PD-02	19	C2S06,19	X1:20	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
04	C2S06,04	X1:05	RELAY 700-HL (24VDC)	JB-402	XV-40205B	Jet Cooker P&ID: 325-01-402-PD-02	20	C2S06,20	X1:21	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
05	C2S06,05	X1:06	RELAY 700-HL (24VDC)	JB-402	XV-40205C	Jet Cooker P&ID: 325-01-402-PD-02	21	C2S06,21	X1:22	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
06	C2S06,06	X1:07	RELAY 700-HL (24VDC)	JB-402	XV-40205D	Jet Cooker P&ID: 325-01-402-PD-02	22	C2S06,22	X1:23	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
07	C2S06,07	X1:08	RELAY 700-HL (24VDC)	JB-402	XV-40205F	Jet Cooker P&ID: 325-01-402-PD-02	23	C2S06,23	X1:24	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
08	C2S06,08	X1:09	RELAY 700-HL (24VDC)	JB-402	XV-40222	Jet Cooker P&ID: 325-01-402-PD-02	24	C2S06,24	X1:25	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
09	C2S06,09	X1:10	RELAY 700-HL (24VDC)	JB-402	SPARE	<i>Spare I/O Channel</i>	25	C2S06,25	X1:26	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
10	C2S06,10	X1:11	RELAY 700-HL (24VDC)	JB-402	SPARE	<i>Spare I/O Channel</i>	26	C2S06,26	X1:27	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
11	C2S06,11	X1:12	RELAY 700-HL (24VDC)	JB-402	SPARE	<i>Spare I/O Channel</i>	27	C2S06,27	X1:28	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
12	C2S06,12	X1:13	RELAY 700-HL (24VDC)	JB-402	SPARE	<i>Spare I/O Channel</i>	28	C2S06,28	X1:29	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
13	C2S06,13	X1:14	RELAY 700-HL (24VDC)	JB-402	SPARE	<i>Spare I/O Channel</i>	29	C2S06,29	X1:30	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
14	C2S06,14	X1:15	RELAY 700-HL (24VDC)	JB-402	SPARE	<i>Spare I/O Channel</i>	30	C2S06,30	X1:31	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
15	C2S06,15	X1:16	RELAY 700-HL (24VDC)	JB-402	SPARE	<i>Spare I/O Channel</i>	31	C2S06,31	X1:32	RELAY 700-HL (24VDC)	JB-602	SPARE	<i>Spare I/O Channel</i>														
16	C2S06,16	X1:17	RELAY 700-HL (24VDC)	JB-602	XS-60240B	Fan Command for BGHS-21 (Bag House Air P.. P&ID: 325-01-600-PD-05		C2S06,00	X1:--	RELAY 700-HL (24VDC)	JB-402	XV-40203F	Jet Cooker P&ID: 325-01-402-PD-02														

MODULE: 1756-OB32 CHASSIS: C2 SLOT: 7 TYPE: DO TERMINAL STRIP: C2S07-X1 POINTS: 32								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C2S07,01	X1:02		RELAY 700-HL (24VDC)	JB-607	XS-FH-60210C	Command Remote Reset for AHTR-21 P&ID: 325-01-600-PD-06	17	C2S07,17	X1:18		RELAY 700-HL (24VDC)	JB-607	XV-60262FH	Feed Slurry On/Off Valve for Nozzle #8 (.. P&ID: 325-01-600-PD-02
02	C2S07,02	X1:03		RELAY 700-HL (24VDC)	JB-607	XS-FH-60210D	Spare P&ID: 325-01-600-PD-06	18	C2S07,18	X1:19		RELAY 700-HL (24VDC)	JB-607	XV-60262SE	Steam On/Off Valve for Nozzle #5 (ATVM-1.. P&ID: 325-01-600-PD-02
03	C2S07,03	X1:04		RELAY 700-HL (24VDC)	JB-607	XV-60201	Steam feed to Spray dryer P&ID: 325-01-600-PD-02	19	C2S07,19	X1:20		RELAY 700-HL (24VDC)	JB-607	XV-60262SF	Steam On/Off Valve for Nozzle #6 (ATVM-1.. P&ID: 325-01-600-PD-02
04	C2S07,04	X1:05		RELAY 700-HL (24VDC)	JB-607	XV-60261	Instrument XV-60261 P&ID: 325-01-600-PD-02	20	C2S07,20	X1:21		RELAY 700-HL (24VDC)	JB-607	XV-60262SG	Steam On/Off Valve for Nozzle #7 (ATVM-1.. P&ID: 325-01-600-PD-02
05	C2S07,05	X1:06		RELAY 700-HL (24VDC)	JB-607	XV-60261FA	Feed Slurry On/Off Valve for Nozzle #1 (.. P&ID: 325-01-600-PD-02	21	C2S07,21	X1:22		RELAY 700-HL (24VDC)	JB-607	XV-60262SH	Steam On/Off Valve for Nozzle #8 (ATVM-1.. P&ID: 325-01-600-PD-02
06	C2S07,06	X1:07		RELAY 700-HL (24VDC)	JB-607	XV-60261FB	Feed Slurry On/Off Valve for Nozzle #2 (.. P&ID: 325-01-600-PD-02	22	C2S07,22	X1:23		RELAY 700-HL (24VDC)	JB-607	XV-60263	Instrument XV-60263 P&ID: 325-01-600-PD-02
07	C2S07,07	X1:08		RELAY 700-HL (24VDC)	JB-607	XV-60261FC	Feed Slurry On/Off Valve for Nozzle #3 (.. P&ID: 325-01-600-PD-02	23	C2S07,23	X1:24		RELAY 700-HL (24VDC)	JB-607	XV-60263FJ	Feed Slurry On/Off Valve for Nozzle #9 (.. P&ID: 325-01-600-PD-02
08	C2S07,08	X1:09		RELAY 700-HL (24VDC)	JB-607	XV-60261FD	Feed Slurry On/Off Valve for Nozzle #4 (.. P&ID: 325-01-600-PD-02	24	C2S07,24	X1:25		RELAY 700-HL (24VDC)	JB-607	XV-60263FK	Feed Slurry On/Off Valve for Nozzle #10.. P&ID: 325-01-600-PD-02
09	C2S07,09	X1:10		RELAY 700-HL (24VDC)	JB-607	XV-60261SA	Steam On/Off Valve for Nozzle #1 (ATVM-1.. P&ID: 325-01-600-PD-02	25	C2S07,25	X1:26		RELAY 700-HL (24VDC)	JB-607	XV-60263FL	Feed Slurry On/Off Valve for Nozzle #11.. P&ID: 325-01-600-PD-02
10	C2S07,10	X1:11		RELAY 700-HL (24VDC)	JB-607	XV-60261SB	Steam On/Off Valve for Nozzle #2 (ATVM-1.. P&ID: 325-01-600-PD-02	26	C2S07,26	X1:27		RELAY 700-HL (24VDC)	JB-607	XV-60263FM	Feed Slurry On/Off Valve for Nozzle #12.. P&ID: 325-01-600-PD-02
11	C2S07,11	X1:12		RELAY 700-HL (24VDC)	JB-607	XV-60261SC	Steam On/Off Valve for Nozzle #3 (ATVM-1.. P&ID: 325-01-600-PD-02	27	C2S07,27	X1:28		RELAY 700-HL (24VDC)	JB-607	XV-60263SJ	Steam On/Off Valve for Nozzle #9 (ATVM-1.. P&ID: 325-01-600-PD-02
12	C2S07,12	X1:13		RELAY 700-HL (24VDC)	JB-607	XV-60261SD	Steam On/Off Valve for Nozzle #4 (ATVM-1.. P&ID: 325-01-600-PD-02	28	C2S07,28	X1:29		RELAY 700-HL (24VDC)	JB-607	XV-60263SK	Steam On/Off Valve for Nozzle #10 (ATVM-.. P&ID: 325-01-600-PD-02
13	C2S07,13	X1:14		RELAY 700-HL (24VDC)	JB-607	XV-60262	Instrument XV-60262 P&ID: 325-01-600-PD-02	29	C2S07,29	X1:30		RELAY 700-HL (24VDC)	JB-607	XV-60263SL	Steam On/Off Valve for Nozzle #11 (ATVM-.. P&ID: 325-01-600-PD-02
14	C2S07,14	X1:15		RELAY 700-HL (24VDC)	JB-607	XV-60262FE	Feed Slurry On/Off Valve for Nozzle #5 (.. P&ID: 325-01-600-PD-02	30	C2S07,30	X1:31		RELAY 700-HL (24VDC)	JB-607	XV-60263SM	Steam On/Off Valve for Nozzle #12 (ATVM-.. P&ID: 325-01-600-PD-02
15	C2S07,15	X1:16		RELAY 700-HL (24VDC)	JB-607	XV-60262FF	Feed Slurry On/Off Valve for Nozzle #6 (.. P&ID: 325-01-600-PD-02	31	C2S07,31	X1:32		RELAY 700-HL (24VDC)	JB-607	SPARE	Spare I/O Channel
16	C2S07,16	X1:17		RELAY 700-HL (24VDC)	JB-607	XV-60262FG	Feed Slurry On/Off Valve for Nozzle #7 (.. P&ID: 325-01-600-PD-02		C2S07,00	X1:--		RELAY 700-HL (24VDC)	JB-607	SV-60207	Instr. Air On/Off Valve for Camera #4 (A.. P&ID: 325-01-600-PD-07
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT												AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION			
CHASSIS C2 SLOT 7 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756												DWG NO: KAL-JC-WIR-20015 REV: 3.6 DATE: 2026-09-08			
DRAWN: AEC-ENG CHECKED: LEAD-PE						STATUS: APPROVED SCALE: N.T.S.						PROJECT NO: R5THS00138-JC		SHEET NO: 18 / 40	

MODULE: 1756-OB32 CHASSIS: C2 SLOT: 8 TYPE: DO TERMINAL STRIP: C2S08-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL															
PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01							C2S08,01	X1:02		RELAY 700-HL (24VDC)		JB-607	XV-60264FN	Feed Slurry On/Off Valve for Nozzle #13 ... P&ID: 325-01-600-PD-02		17							C2S08,17	X1:18		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
02							C2S08,02	X1:03		RELAY 700-HL (24VDC)		JB-607	XV-60264FP	Feed Slurry On/Off Valve for Nozzle #14 ... P&ID: 325-01-600-PD-02		18							C2S08,18	X1:19		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
03							C2S08,03	X1:04		RELAY 700-HL (24VDC)		JB-607	XV-60264FQ	Feed Slurry On/Off Valve for Nozzle #15 ... P&ID: 325-01-600-PD-02		19							C2S08,19	X1:20		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
04							C2S08,04	X1:05		RELAY 700-HL (24VDC)		JB-607	XV-60264FR	Feed Slurry On/Off Valve for Nozzle #16 ... P&ID: 325-01-600-PD-02		20							C2S08,20	X1:21		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
05							C2S08,05	X1:06		RELAY 700-HL (24VDC)		JB-607	XV-60264SN	Steam On/Off Valve for Nozzle #13 (ATVM-... P&ID: 325-01-600-PD-02		21							C2S08,21	X1:22		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
06							C2S08,06	X1:07		RELAY 700-HL (24VDC)		JB-607	XV-60264SP	Steam On/Off Valve for Nozzle #14 (ATVM-... P&ID: 325-01-600-PD-02		22							C2S08,22	X1:23		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
07							C2S08,07	X1:08		RELAY 700-HL (24VDC)		JB-607	XV-60264SQ	Steam On/Off Valve for Nozzle #15 (ATVM-... P&ID: 325-01-600-PD-02		23							C2S08,23	X1:24		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
08							C2S08,08	X1:09		RELAY 700-HL (24VDC)		JB-607	XV-60264SR	Steam On/Off Valve for Nozzle #16 (ATVM-... P&ID: 325-01-600-PD-02		24							C2S08,24	X1:25		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
09							C2S08,09	X1:10		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		25							C2S08,25	X1:26		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
10							C2S08,10	X1:11		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		26							C2S08,26	X1:27		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
11							C2S08,11	X1:12		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		27							C2S08,27	X1:28		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
12							C2S08,12	X1:13		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		28							C2S08,28	X1:29		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
13							C2S08,13	X1:14		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		29							C2S08,29	X1:30		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
14							C2S08,14	X1:15		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		30							C2S08,30	X1:31		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
15							C2S08,15	X1:16		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		31							C2S08,31	X1:32		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel	
16							C2S08,16	X1:17		RELAY 700-HL (24VDC)		JB-607	SPARE	Spare I/O Channel		C2S08,00							X1:--		RELAY 700-HL (24VDC)		JB-607	XV-60264	Instrument XV-60264 P&ID: 325-01-600-PD-02		
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT																AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION															
CHASSIS C2 SLOT 8 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756																DWG NO: KAL-JC-WIR-20016 REV: 3.6 DATE: 2026-09-08															
DRAWN: AEC-ENG CHECKED: LEAD-PE								STATUS: APPROVED SCALE: N.T.S.								PROJECT NO: R5THS00138-JC				SHEET NO: 19 / 40											

MODULE: 1756-IF16 CHASSIS: C2 SLOT: 9 TYPE: AI TERMINAL STRIP: C2S09-X1 POINTS: 16																RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL															
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION																								
01	C2S09,01	X1:02		LOOP POWERED (2-W)	JB-402	FT-40221	Jet Cooker P&ID: 325-01-402-PD-02																								
02	C2S09,02	X1:03		LOOP POWERED (2-W)	JB-402	FT-40222	Jet Cooker P&ID: 325-01-402-PD-02																								
03	C2S09,03	X1:04		LOOP POWERED (2-W)	JB-402	FT-40222	Jet Cooker P&ID: 325-01-402-PD-02																								
04	C2S09,04	X1:05		LOOP POWERED (2-W)	JB-402	PT-40222	Jet Cooker P&ID: 325-01-402-PD-02																								
05	C2S09,05	X1:06		LOOP POWERED (2-W)	JB-402	TT-40221	Jet Cooker P&ID: 325-01-402-PD-02																								
06	C2S09,06	X1:07		LOOP POWERED (2-W)	JB-402	TT-40222	Jet Cooker P&ID: 325-01-402-PD-02																								
07	C2S09,07	X1:08		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
08	C2S09,08	X1:09		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
09	C2S09,09	X1:10		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
10	C2S09,10	X1:11		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
11	C2S09,11	X1:12		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
12	C2S09,12	X1:13		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
13	C2S09,13	X1:14		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
14	C2S09,14	X1:15		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
15	C2S09,15	X1:16		LOOP POWERED (2-W)	JB-402	SPARE	Spare I/O Channel																								
	C2S09,00	X1:--		LOOP POWERED (2-W)	JB-402	FT-40221	Jet Cooker P&ID: 325-01-402-PD-02																								
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT																AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION															
CHASSIS C2 SLOT 9 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756																DWG NO: KAL-JC-WIR-20017 REV: 3.6 DATE: 2026-09-08															
DRAWN: AEC-ENG CHECKED: LEAD-PE										STATUS: APPROVED SCALE: N.T.S.										PROJECT NO: R5THS00138-JC					SHEET NO: 20 / 40						

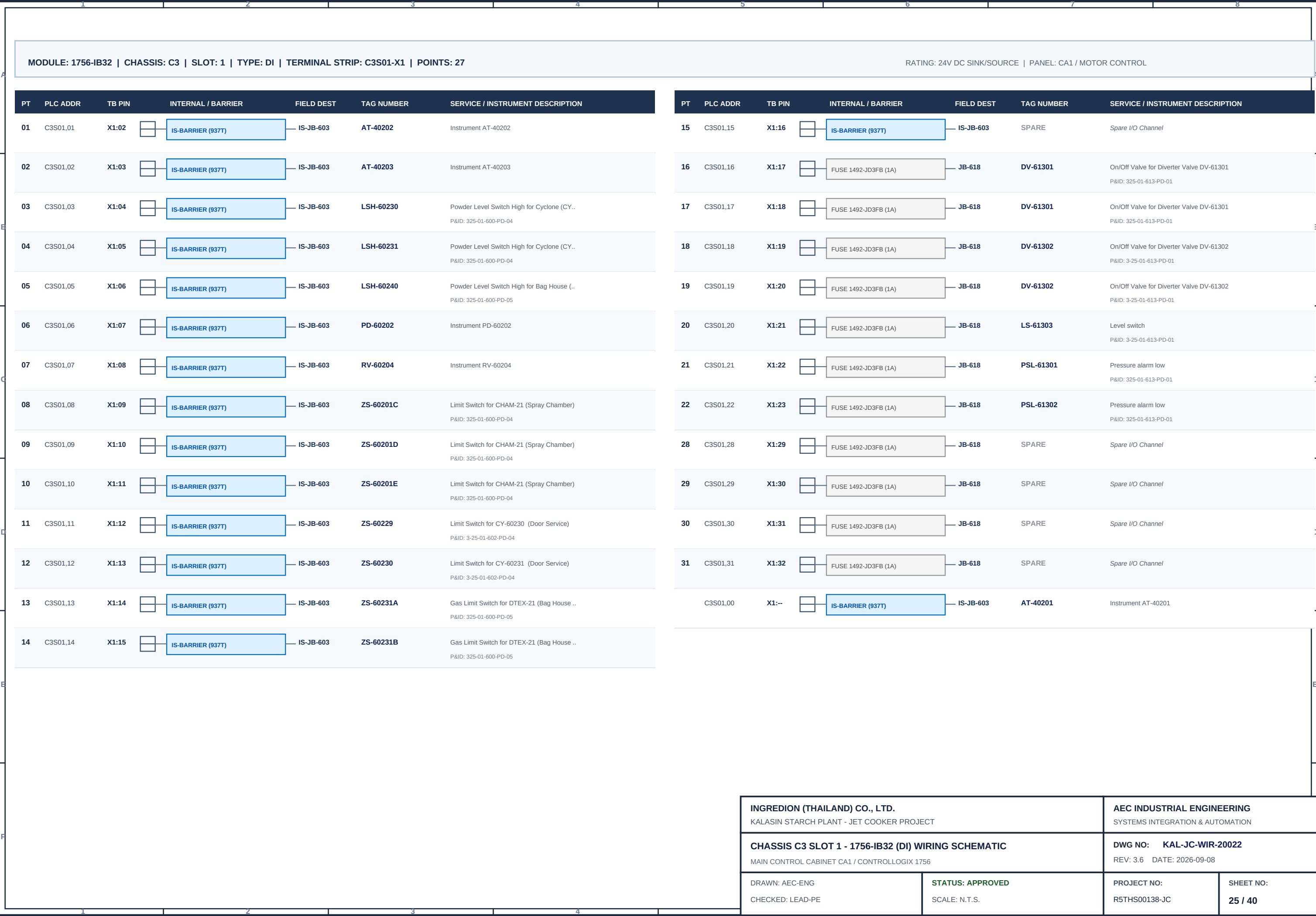
MODULE: 1756-IF16 CHASSIS: C2 SLOT: 10 TYPE: AI TERMINAL STRIP: C2S10-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C2S10,01	X1:02		LOOP POWERED (2-W)	JB-602	DPT-60203	Gas Pressure Differential Transmitter f.. P&ID: 325-01-600-PD-03	
02	C2S10,02	X1:03		LOOP POWERED (2-W)	JB-602	FT-60202	Steam feed to Spray dryer P&ID: 325-01-600-PD-09	
03	C2S10,03	X1:04		LOOP POWERED (2-W)	JB-602	PT-60210	Gas Pressure Transmitter for Hot Air Duc.. P&ID: 325-01-600-PD-03	
04	C2S10,04	X1:05		LOOP POWERED (2-W)	JB-602	PT-60215	15% Polypropylene Glycol Pressure Indica.. P&ID: 325-01-600-PD-03	
05	C2S10,05	X1:06		LOOP POWERED (2-W)	JB-602	PT-60216	Gas Pressure Transmitter for Combustion .. P&ID: 325-01-600-PD-03	
06	C2S10,06	X1:07		LOOP POWERED (2-W)	JB-602	PT-60240	Gas Pressure Transmitter for Bag House E.. P&ID: 325-01-600-PD-05	
07	C2S10,07	X1:08		LOOP POWERED (2-W)	JB-602	PT-60241	Gas Pressure Transmitter for Exhaust Fan.. P&ID: 325-01-600-PD-05	
08	C2S10,08	X1:09		LOOP POWERED (2-W)	JB-602	TT-60210	Gas Temperature Element for Hot Air Duct.. P&ID: 325-01-600-PD-03	
09	C2S10,09	X1:10		LOOP POWERED (2-W)	JB-602	TT-60210A	Instrument TT-60210A P&ID: 325-01-600-PD-03	
10	C2S10,10	X1:11		LOOP POWERED (2-W)	JB-602	TT-60210B	Instrument TT-60210B P&ID: 325-01-600-PD-03	
11	C2S10,11	X1:12		LOOP POWERED (2-W)	JB-602	TT-60211	Gas Temperature Element for Hot Air Duct.. P&ID: 325-01-600-PD-03	
12	C2S10,12	X1:13		LOOP POWERED (2-W)	JB-602	TT-60212A	Bearing Temperature Transmitter for Main.. P&ID: 325-01-600-PD-03	
13	C2S10,13	X1:14		LOOP POWERED (2-W)	JB-602	TT-60212B	Bearing Temperature Transmitter for Main.. P&ID: 325-01-600-PD-03	
14	C2S10,14	X1:15		LOOP POWERED (2-W)	JB-602	TT-60214	Steam Temperature Transmitter for Cold A.. P&ID: 325-01-600-PD-03	
15	C2S10,15	X1:16		LOOP POWERED (2-W)	JB-602	TT-60215A	Liquid Temperature Transmitter for Heat .. P&ID: 325-01-600-PD-03	
	C2S10,00	X1:--		LOOP POWERED (2-W)	JB-602	AT-60241	Dust Analyzer Transmitter for Exhaust Du.. P&ID: 325-01-600-PD-05	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C2 SLOT 10 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20018 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 21 / 40

MODULE: 1756-IF16 CHASSIS: C2 SLOT: 12 TYPE: AI TERMINAL STRIP: C2S12-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C2S12,01	X1:02		LOOP POWERED (2-W)	JB-607	FT-60221	Cool air Flow P&ID: 325-01-600-PD-04	
02	C2S12,02	X1:03		LOOP POWERED (2-W)	JB-607	FT-60261FA	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
03	C2S12,03	X1:04		LOOP POWERED (2-W)	JB-607	FT-60261FB	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
04	C2S12,04	X1:05		LOOP POWERED (2-W)	JB-607	FT-60261FC	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
05	C2S12,05	X1:06		LOOP POWERED (2-W)	JB-607	FT-60261FD	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
06	C2S12,06	X1:07		LOOP POWERED (2-W)	JB-607	FT-60262FE	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
07	C2S12,07	X1:08		LOOP POWERED (2-W)	JB-607	FT-60262FF	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
08	C2S12,08	X1:09		LOOP POWERED (2-W)	JB-607	FT-60262FG	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
09	C2S12,09	X1:10		LOOP POWERED (2-W)	JB-607	FT-60262FH	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
10	C2S12,10	X1:11		LOOP POWERED (2-W)	JB-607	FT-60263FJ	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
11	C2S12,11	X1:12		LOOP POWERED (2-W)	JB-607	FT-60263FK	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
12	C2S12,12	X1:13		LOOP POWERED (2-W)	JB-607	FT-60263FL	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
13	C2S12,13	X1:14		LOOP POWERED (2-W)	JB-607	FT-60263FM	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
14	C2S12,14	X1:15		LOOP POWERED (2-W)	JB-607	FT-60264FN	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
15	C2S12,15	X1:16		LOOP POWERED (2-W)	JB-607	FT-60264FP	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
	C2S12,00	X1:--		LOOP POWERED (2-W)	JB-607	FT-60207	Instr. Air Flow Transmitter for Camera #.. P&ID: 325-01-600-PD-07	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C2 SLOT 12 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20020 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 23 / 40

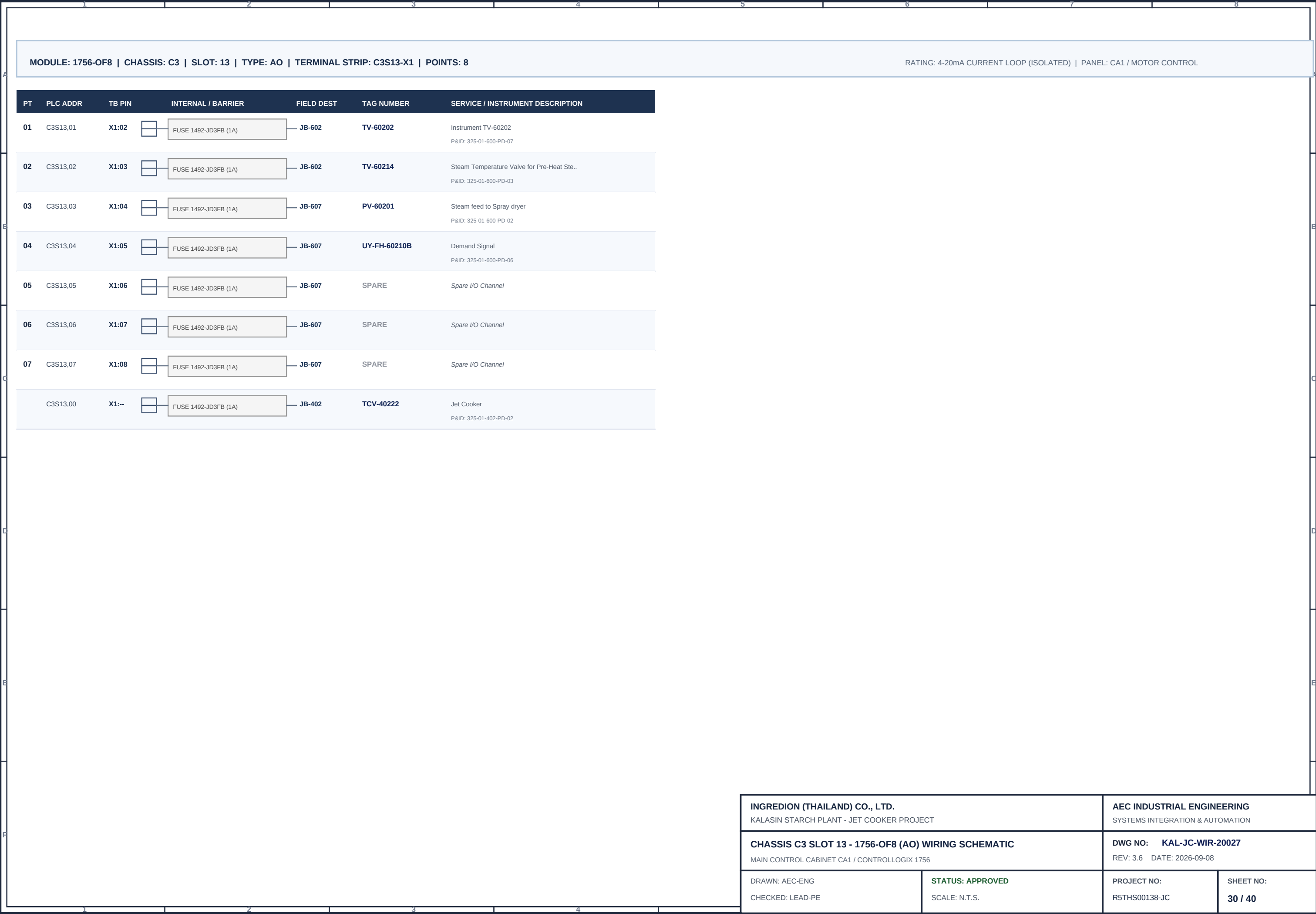
MODULE: 1756-IF16 CHASSIS: C2 SLOT: 13 TYPE: AI TERMINAL STRIP: C2S13-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C2S13,01	X1:02		LOOP POWERED (2-W)	JB-607	FT-60264FR	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
02	C2S13,02	X1:03		LOOP POWERED (2-W)	JB-607	PT-60201	Steam feed to Spray dryer P&ID: 3-25-01-602-PD-02	
03	C2S13,03	X1:04		LOOP POWERED (2-W)	JB-607	PT-60207	Instr. Air Pressure Transmitter for Came.. P&ID: 325-01-600-PD-07	
04	C2S13,04	X1:05		LOOP POWERED (2-W)	JB-607	PT-60221	Gas Pressure Transmitter for Cooling Air.. P&ID: 325-01-600-PD-02	
05	C2S13,05	X1:06		LOOP POWERED (2-W)	JB-607	PT-60221	Gas Pressure Transmitter for Cooling Air.. P&ID: 325-01-600-PD-04	
06	C2S13,06	X1:07		LOOP POWERED (2-W)	JB-607	TI-FH-60210	Gas Burner Firing Rate Temp Signal to AH.. P&ID: 325-01-600-PD-06	
07	C2S13,07	X1:08		LOOP POWERED (2-W)	JB-607	TT-60201	Steam feed to Spray dryer P&ID: 325-01-600-PD-02	
08	C2S13,08	X1:09		LOOP POWERED (2-W)	JB-607	TT-60202	Compress air temp to camera P&ID: 325-01-600-PD-07	
09	C2S13,09	X1:10		LOOP POWERED (2-W)	JB-607	TT-60207	Instr. Air Temperature Transmitter for A.. P&ID: 325-01-600-PD-07	
10	C2S13,10	X1:11		LOOP POWERED (2-W)	JB-607	UY-FH-60210A	Comb fan VFD Speed Control P&ID: 325-01-600-PD-06	
11	C2S13,11	X1:12		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
12	C2S13,12	X1:13		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
13	C2S13,13	X1:14		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
14	C2S13,14	X1:15		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
15	C2S13,15	X1:16		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
	C2S13,00	X1:--		LOOP POWERED (2-W)	JB-607	FT-60264FQ	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT							AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C2 SLOT 13 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756							DWG NO: KAL-JC-WIR-20021 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.			PROJECT NO: R5THS00138-JC	SHEET NO: 24 / 40

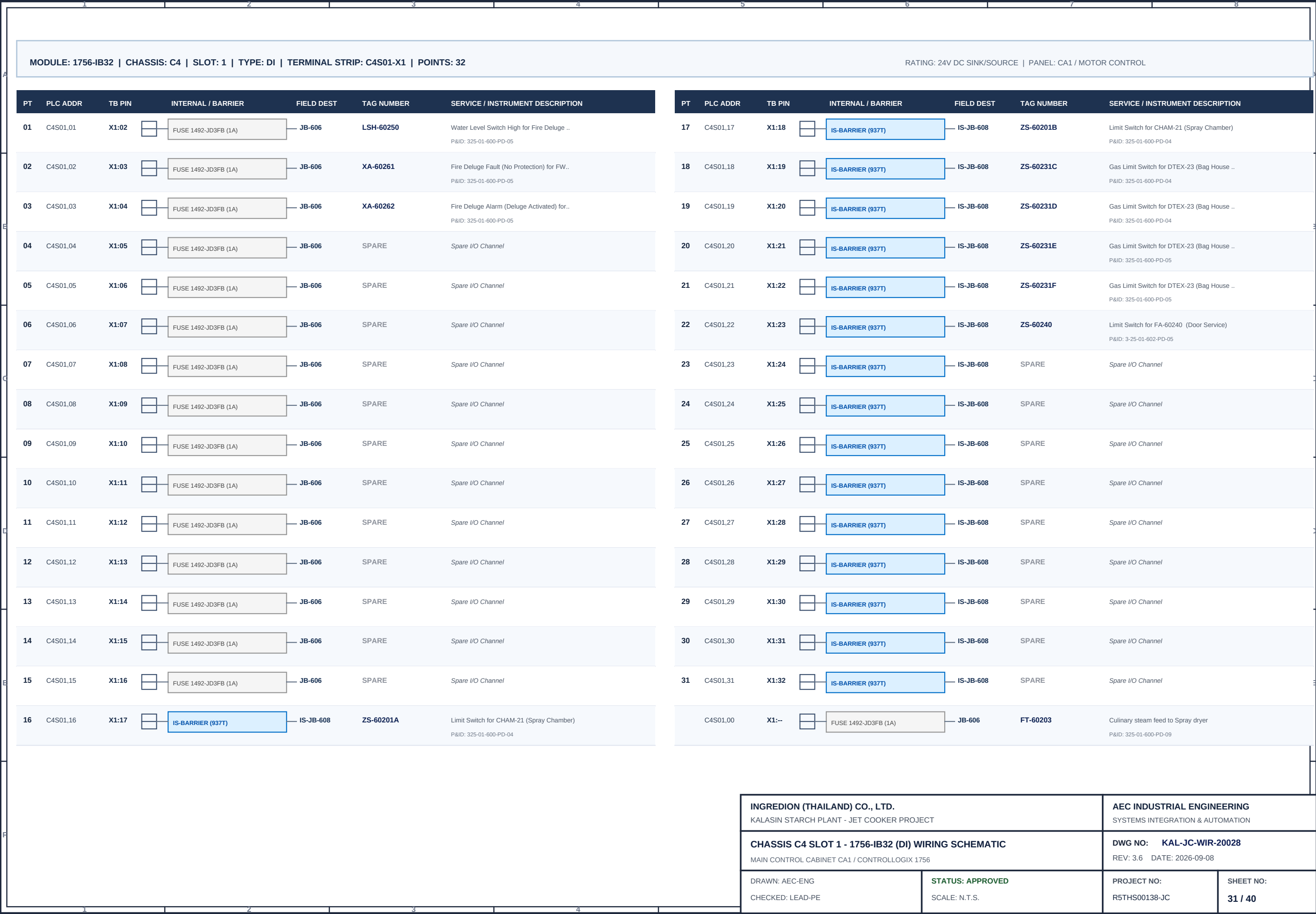


MODULE: 1756-OB32 CHASSIS: C3 SLOT: 3 TYPE: DO TERMINAL STRIP: C3S03-X1 POINTS: 32				RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL			
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C3S03,01	X1:02	RELAY 700-HL (24VDC)	JB-618	DV-61302	On/Off Valve for Diverter Valve DV-61302 P&ID: 3-25-01-613-PD-01	
02	C3S03,02	X1:03	RELAY 700-HL (24VDC)	JB-618	SV-61301	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-01	
03	C3S03,03	X1:04	RELAY 700-HL (24VDC)	JB-618	SV-61302	Air service P&ID: 325-01-613-PD-01	
04	C3S03,04	X1:05	RELAY 700-HL (24VDC)	JB-618	SV-61303	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-01	
05	C3S03,05	X1:06	RELAY 700-HL (24VDC)	JB-618	SV-61304	PA On/Off Valve for Conveying Cooler Cyc.. P&ID: 325-01-613-PD-01	
06	C3S03,06	X1:07	RELAY 700-HL (24VDC)	JB-618	SV-61305	Air service P&ID: 325-01-613-PD-01	
07	C3S03,07	X1:08	RELAY 700-HL (24VDC)	JB-618	SV-61306	Instrument SV-61306 P&ID: 3-25-01-613-PD-01	
08	C3S03,08	X1:09	RELAY 700-HL (24VDC)	JB-618	XV-61301	Product feed-out Packing silo P&ID: 3-25-01-613-PD-01	
09	C3S03,09	X1:10	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
10	C3S03,10	X1:11	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
11	C3S03,11	X1:12	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
12	C3S03,12	X1:13	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
13	C3S03,13	X1:14	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
14	C3S03,14	X1:15	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
15	C3S03,15	X1:16	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
16	C3S03,16	X1:17	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
17	C3S03,17	X1:18	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
18	C3S03,18	X1:19	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
19	C3S03,19	X1:20	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
20	C3S03,20	X1:21	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
21	C3S03,21	X1:22	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
22	C3S03,22	X1:23	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
23	C3S03,23	X1:24	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
24	C3S03,24	X1:25	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
25	C3S03,25	X1:26	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
26	C3S03,26	X1:27	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
27	C3S03,27	X1:28	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
28	C3S03,28	X1:29	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
29	C3S03,29	X1:30	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
30	C3S03,30	X1:31	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
31	C3S03,31	X1:32	RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel	
	C3S03,00	X1:--	RELAY 700-HL (24VDC)	JB-618	DV-61301	On/Off Valve for Diverter Valve DV-61301 P&ID: 325-01-613-PD-01	
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT						AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C3 SLOT 3 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756						DWG NO: KAL-JC-WIR-20024 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE			STATUS: APPROVED SCALE: N.T.S.			PROJECT NO: R5THS00138-JC	SHEET NO: 27 / 40

MODULE: 1756-IF16 CHASSIS: C3 SLOT: 5 TYPE: AI TERMINAL STRIP: C3S05-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C3S05,01	X1:02		LOOP POWERED (2-W)	JB-618	PT-61306	Instrument PT-61306 P&ID: 3-25-01-613-PD-01	
02	C3S05,02	X1:03		LOOP POWERED (2-W)	JB-618	TT-61303	Instrument TT-61303 P&ID: 3-25-01-613-PD-01	
03	C3S05,03	X1:04		LOOP POWERED (2-W)	JB-618	TT-61304	Instrument TT-61304 P&ID: 3-25-01-613-PD-01	
04	C3S05,04	X1:05		IS-TX-BARRIER (937T)	IS-JB-618	DPT-61301	Vent filter pressure P&ID: 325-01-613-PD-01	
05	C3S05,05	X1:06		IS-TX-BARRIER (937T)	IS-JB-618	DPT-61302	Vent filter pressure P&ID: 325-01-613-PD-01	
06	C3S05,06	X1:07		IS-TX-BARRIER (937T)	IS-JB-618	LT-61301	Level powder in silo P&ID: 325-01-613-PD-01	
07	C3S05,07	X1:08		IS-TX-BARRIER (937T)	IS-JB-618	PT-61301	Pressure in silo P&ID: 325-01-613-PD-01	
08	C3S05,08	X1:09		IS-TX-BARRIER (937T)	IS-JB-618	PT-61303	Powder Pressure Transmitter for Cooling .. P&ID: 325-01-613-PD-01	
09	C3S05,09	X1:10		IS-TX-BARRIER (937T)	IS-JB-618	PT-61304	Gas Pressure Transmitter for Storage Bin.. P&ID: 325-01-613-PD-01	
10	C3S05,10	X1:11		IS-TX-BARRIER (937T)	IS-JB-618	PT-61305	Gas Pressure Transmitter for Exhaust Duc.. P&ID: 325-01-613-PD-01	
11	C3S05,11	X1:12		IS-TX-BARRIER (937T)	IS-JB-618	TT-61301	Gas Temperature Transmitter for Cold Air.. P&ID: 325-01-613-PD-01	
12	C3S05,12	X1:13		IS-TX-BARRIER (937T)	IS-JB-618	TT-61302	Gas and Powder Temperature Transmitter f.. P&ID: 325-01-613-PD-01	
13	C3S05,13	X1:14		IS-TX-BARRIER (937T)	IS-JB-618	SPARE	Spare I/O Channel	
14	C3S05,14	X1:15		IS-TX-BARRIER (937T)	IS-JB-618	SPARE	Spare I/O Channel	
15	C3S05,15	X1:16		IS-TX-BARRIER (937T)	IS-JB-618	SPARE	Spare I/O Channel	
	C3S05,00	X1:--		LOOP POWERED (2-W)	JB-618	PT-61303	Powder Pressure Transmitter for Cooling .. P&ID: 3-25-01-613-PD-01	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C3 SLOT 5 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20026 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 29 / 40

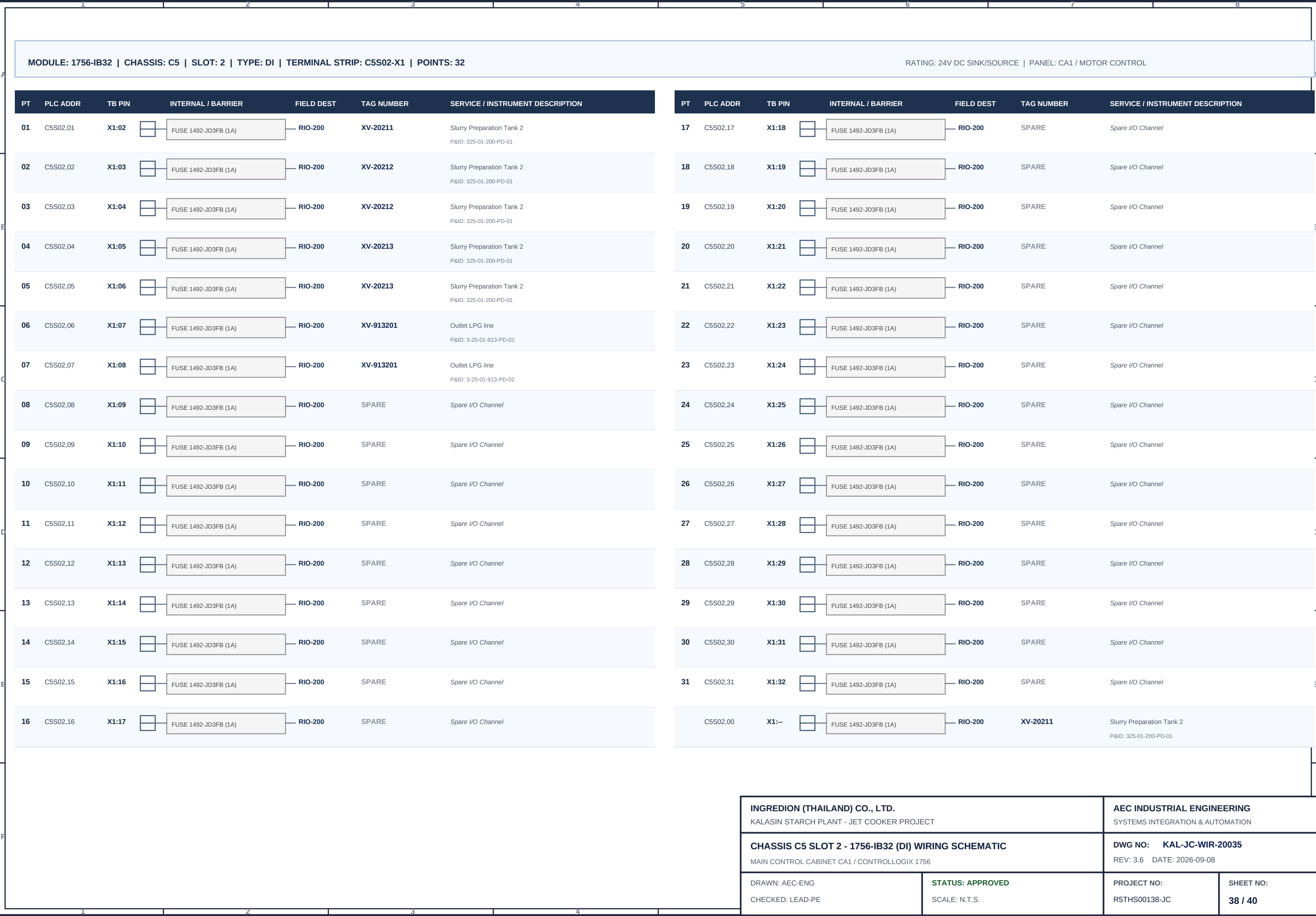


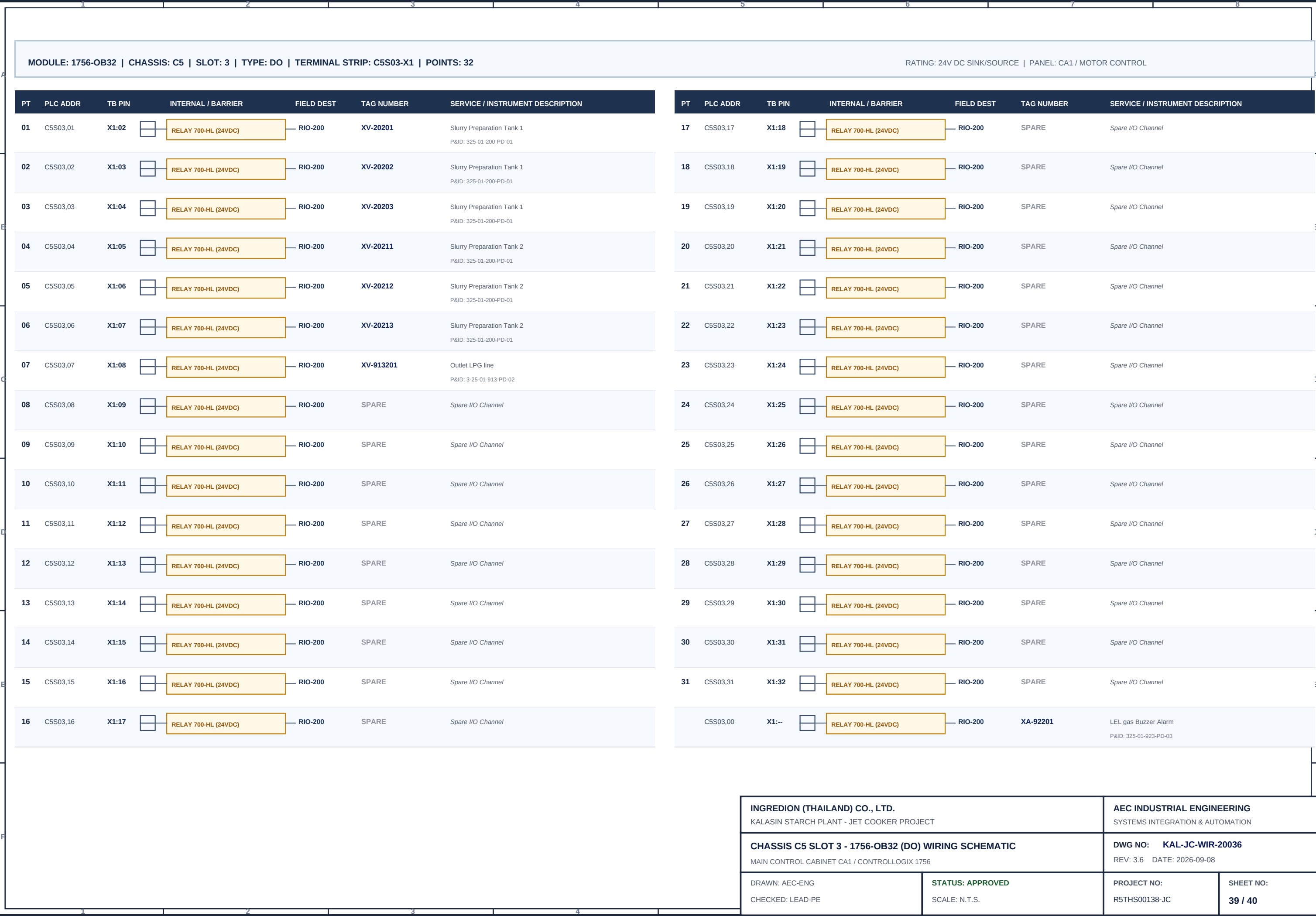


MODULE: 1756-IB32 CHASSIS: C4 SLOT: 2 TYPE: DI TERMINAL STRIP: C4S02-X1 POINTS: 32				RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL			
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C4S02,01	X1:02	FUSE 1492-JD3FB (1A)	JB-612	PSL-61306	Pressure alarm low P&ID: 325-01-613-PD-05	
02	C4S02,02	X1:03	FUSE 1492-JD3FB (1A)	JB-612	PVM-61301	Start / Stop Command for FVNR PV-61301 (...) P&ID: 325-01-613-PD-03	
03	C4S02,03	X1:04	FUSE 1492-JD3FB (1A)	JB-612	PVM-61301	Start / Stop Command for FVNR PV-61301 (...) P&ID: 325-01-613-PD-03	
04	C4S02,04	X1:05	FUSE 1492-JD3FB (1A)	JB-612	XA-61301A	Request filter plant P&ID: 325-01-613-PD-02	
05	C4S02,05	X1:06	FUSE 1492-JD3FB (1A)	JB-612	XA-61301B	Request material transport P&ID: 325-01-613-PD-02	
06	C4S02,06	X1:07	FUSE 1492-JD3FB (1A)	JB-612	XA-61301C	Request belt plant P&ID: 325-01-613-PD-02	
07	C4S02,07	X1:08	FUSE 1492-JD3FB (1A)	JB-612	XA-61301D	Machine in operation P&ID: 325-01-613-PD-02	
08	C4S02,08	X1:09	FUSE 1492-JD3FB (1A)	JB-612	XA-61301E	General Fault P&ID: 325-01-613-PD-02	
09	C4S02,09	X1:10	FUSE 1492-JD3FB (1A)	JB-612	XA-61301F	Machine is ready P&ID: 325-01-613-PD-02	
10	C4S02,10	X1:11	FUSE 1492-JD3FB (1A)	JB-612	XA-61301G	Emergency stop activated P&ID: 325-01-613-PD-02	
11	C4S02,11	X1:12	FUSE 1492-JD3FB (1A)	JB-612	XA-61301H	Silo minimum P&ID: 325-01-613-PD-02	
12	C4S02,12	X1:13	FUSE 1492-JD3FB (1A)	JB-612	XA-61301J	Silo maximum P&ID: 325-01-613-PD-02	
13	C4S02,13	X1:14	FUSE 1492-JD3FB (1A)	JB-612	XV-61302	Dust collector feed vent P&ID: 325-01-613-PD-02	
14	C4S02,14	X1:15	FUSE 1492-JD3FB (1A)	JB-612	XV-61302	Dust collector feed vent P&ID: 325-01-613-PD-02	
15	C4S02,15	X1:16	FUSE 1492-JD3FB (1A)	JB-612	XV-61303	Dust collector feed from packer P&ID: 325-01-613-PD-02	
16	C4S02,16	X1:17	FUSE 1492-JD3FB (1A)	JB-612	XV-61303	Dust collector feed from packer P&ID: 325-01-613-PD-02	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
17	C4S02,17	X1:18	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
18	C4S02,18	X1:19	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
19	C4S02,19	X1:20	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
20	C4S02,20	X1:21	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
21	C4S02,21	X1:22	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
22	C4S02,22	X1:23	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
23	C4S02,23	X1:24	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
24	C4S02,24	X1:25	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
25	C4S02,25	X1:26	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
26	C4S02,26	X1:27	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
27	C4S02,27	X1:28	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
28	C4S02,28	X1:29	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
29	C4S02,29	X1:30	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
30	C4S02,30	X1:31	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
31	C4S02,31	X1:32	FUSE 1492-JD3FB (1A)	JB-612	SPARE	Spare I/O Channel	
	C4S02,00	X1:--	FUSE 1492-JD3FB (1A)	JB-612	PSL-61305	Pressure alarm low P&ID: 325-01-613-PD-03	
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT						AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C4 SLOT 2 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756						DWG NO: KAL-JC-WIR-20029 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE			STATUS: APPROVED SCALE: N.T.S.			PROJECT NO: R5THS00138-JC	SHEET NO: 32 / 40

MODULE: 1756-IB32 CHASSIS: C4 SLOT: 3 TYPE: DI TERMINAL STRIP: C4S03-X1 POINTS: 31								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C4S03,01	X1:02		IS-BARRIER (937T)	IS-JB-612	ISV-61304	Explosion isolation flap valve ISV-61304 P&ID: 325-01-613-PD-05	18	C4S03,18	X1:19		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
02	C4S03,02	X1:03		IS-BARRIER (937T)	IS-JB-612	LSH-61304	Big bag level alarm high P&ID: 325-01-613-PD-03	19	C4S03,19	X1:20		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
03	C4S03,03	X1:04		IS-BARRIER (937T)	IS-JB-612	LSH-61305	Big bag level alarm high P&ID: 325-01-613-PD-05	20	C4S03,20	X1:21		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
04	C4S03,04	X1:05		IS-BARRIER (937T)	IS-JB-612	LSH-61307	High Level alarm reject screen P&ID: 3-25-01-613-PD-01	21	C4S03,21	X1:22		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
05	C4S03,05	X1:06		IS-BARRIER (937T)	IS-JB-612	LSH-61312	Level alarm high P&ID: 325-01-613-PD-05	22	C4S03,22	X1:23		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
06	C4S03,06	X1:07		IS-BARRIER (937T)	IS-JB-612	PSE-61302	Explosion Vent PSE-61302 P&ID: 325-01-613-PD-03	23	C4S03,23	X1:24		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
07	C4S03,07	X1:08		IS-BARRIER (937T)	IS-JB-612	PSE-61303	Explosion Vent PSE-61303 P&ID: 325-01-613-PD-05	24	C4S03,24	X1:25		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
09	C4S03,09	X1:10		IS-BARRIER (937T)	IS-JB-612	RV-61301	Instrument RV-61301	25	C4S03,25	X1:26		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
10	C4S03,10	X1:11		IS-BARRIER (937T)	IS-JB-612	RV-61304	Instrument RV-61304	26	C4S03,26	X1:27		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
11	C4S03,11	X1:12		IS-BARRIER (937T)	IS-JB-612	RV-61305	Instrument RV-61305	27	C4S03,27	X1:28		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
12	C4S03,12	X1:13		IS-BARRIER (937T)	IS-JB-612	SC-61301	Instrument SC-61301	28	C4S03,28	X1:29		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
13	C4S03,13	X1:14		IS-BARRIER (937T)	IS-JB-612	SC-61302	Instrument SC-61302	29	C4S03,29	X1:30		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
14	C4S03,14	X1:15		IS-BARRIER (937T)	IS-JB-612	ZS-61312	Position Magnet Door P&ID: 325-01-613-PD-01	30	C4S03,30	X1:31		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
15	C4S03,15	X1:16		IS-BARRIER (937T)	IS-JB-612	ZS-61313	Position Magnet Door P&ID: 325-01-613-PD-01	31	C4S03,31	X1:32		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
16	C4S03,16	X1:17		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel		C4S03,00	X1:--		IS-BARRIER (937T)	IS-JB-612	ISV-61303	Explosion isolation flap valve ISV-61303 P&ID: 325-01-613-PD-03
17	C4S03,17	X1:18		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel								
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT								AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION							
CHASSIS C4 SLOT 3 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756								DWG NO: KAL-JC-WIR-20030 REV: 3.6 DATE: 2026-09-08							
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.				PROJECT NO: R5THS00138-JC		SHEET NO: 33 / 40					

MODULE: 1756-OB32 CHASSIS: C4 SLOT: 4 TYPE: DO TERMINAL STRIP: C4S04-X1 POINTS: 32								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C4S04,01	X1:02		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	17	C4S04,17	X1:18		RELAY 700-HL (24VDC)	JB-612	PVM-61301	Start / Stop Command for FVNR PV-61301 (.. P&ID: 325-01-613-PD-03
02	C4S04,02	X1:03		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	18	C4S04,18	X1:19		RELAY 700-HL (24VDC)	JB-612	SV-61307	Instrument SV-61307 P&ID: 3-25-01-613-PD-01
03	C4S04,03	X1:04		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	19	C4S04,19	X1:20		RELAY 700-HL (24VDC)	JB-612	SV-61308	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-03
04	C4S04,04	X1:05		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	20	C4S04,20	X1:21		RELAY 700-HL (24VDC)	JB-612	SV-61309	Air service P&ID: 325-01-613-PD-03
05	C4S04,05	X1:06		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	21	C4S04,21	X1:22		RELAY 700-HL (24VDC)	JB-612	SV-61310	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-05
06	C4S04,06	X1:07		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	22	C4S04,22	X1:23		RELAY 700-HL (24VDC)	JB-612	SV-61311	Air service P&ID: 325-01-613-PD-05
07	C4S04,07	X1:08		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	23	C4S04,23	X1:24		RELAY 700-HL (24VDC)	JB-612	XS-61301A	Filter plant in operation P&ID: 325-01-613-PD-02
08	C4S04,08	X1:09		IS-TX-BARRIER (937T)	IS-JB-608	XA-60201	LEL gas Buzzer Alarm P&ID: 325-01-600-PD-03	24	C4S04,24	X1:25		RELAY 700-HL (24VDC)	JB-612	XS-61301B	Material transport in operation P&ID: 325-01-613-PD-02
09	C4S04,09	X1:10		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	25	C4S04,25	X1:26		RELAY 700-HL (24VDC)	JB-612	XS-61301C	Belt plant in operation P&ID: 325-01-613-PD-02
10	C4S04,10	X1:11		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	26	C4S04,26	X1:27		RELAY 700-HL (24VDC)	JB-612	XS-61301D	Sequence Stop P&ID: 325-01-613-PD-02
11	C4S04,11	X1:12		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	27	C4S04,27	X1:28		RELAY 700-HL (24VDC)	JB-612	XS-61301E	Status from Customer Emergency stop P&ID: 325-01-613-PD-02
12	C4S04,12	X1:13		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	28	C4S04,28	X1:29		RELAY 700-HL (24VDC)	JB-612	XS-61301F	Operation Release P&ID: 325-01-613-PD-02
13	C4S04,13	X1:14		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	29	C4S04,29	X1:30		RELAY 700-HL (24VDC)	JB-612	XV-61302	Dust collector feed vent P&ID: 325-01-613-PD-02
14	C4S04,14	X1:15		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	30	C4S04,30	X1:31		RELAY 700-HL (24VDC)	JB-612	XV-61303	Dust collector feed from packer P&ID: 325-01-613-PD-02
15	C4S04,15	X1:16		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	31	C4S04,31	X1:32		RELAY 700-HL (24VDC)	JB-612	SPARE	Spare I/O Channel
16	C4S04,16	X1:17		RELAY 700-HL (24VDC)	JB-612	PVM-61301	Start / Stop Command for FVNR PV-61301 (.. P&ID: 325-01-613-PD-03		C4S04,00	X1:--		RELAY 700-HL (24VDC)	JB-606	SV-60220	PA On/Off Valve for Chamber PSE purge ai.. P&ID: 325-01-600-PD-04
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT												AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION			
CHASSIS C4 SLOT 4 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756												DWG NO: KAL-JC-WIR-20031 REV: 3.6 DATE: 2026-09-08			
DRAWN: AEC-ENG CHECKED: LEAD-PE						STATUS: APPROVED SCALE: N.T.S.						PROJECT NO: R5THS00138-JC		SHEET NO: 34 / 40	





MODULE: 1756-IF16 CHASSIS: C5 SLOT: 4 TYPE: AI TERMINAL STRIP: C5S04-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C5S04,01	X1:02		LOOP POWERED (2-W)	RIO-200	FT-913201	Flow outlet LPG line P&ID: 325-01-803-PD-02	
02	C5S04,02	X1:03		LOOP POWERED (2-W)	RIO-200	FT-913201	Flow outlet LPG line P&ID: 325-01-803-PD-02	
03	C5S04,03	X1:04		LOOP POWERED (2-W)	RIO-200	FT-936201	Process Water P&ID: 325-01-930-PD-01	
04	C5S04,04	X1:05		LOOP POWERED (2-W)	RIO-200	FT-936202	Process Water P&ID: 325-01-930-PD-02	
05	C5S04,05	X1:06		LOOP POWERED (2-W)	RIO-200	LT-913201	Level LPG in storage P&ID: 325-01-803-PD-01	
06	C5S04,06	X1:07		LOOP POWERED (2-W)	RIO-200	PT-913201	Pressure outlet LPG line P&ID: 325-01-803-PD-02	
07	C5S04,07	X1:08		LOOP POWERED (2-W)	RIO-200	XT-92201	LEL gas analyzer P&ID: 325-01-923-PD-03	
08	C5S04,08	X1:09		LOOP POWERED (2-W)	RIO-200	XV-913201	Outlet LPG line P&ID: 3-25-01-913-PD-02	
09	C5S04,09	X1:10		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
10	C5S04,10	X1:11		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
11	C5S04,11	X1:12		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
12	C5S04,12	X1:13		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
13	C5S04,13	X1:14		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
14	C5S04,14	X1:15		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel	
15	C5S04,15	X1:16		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel	
	C5S04,00	X1:--		LOOP POWERED (2-W)	RIO-200	EXT-BIILER	Instrument EXT-BIILER P&ID: External Remote Setpoint	
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT							AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C5 SLOT 4 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756							DWG NO: KAL-JC-WIR-20037 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.			PROJECT NO: R5THS00138-JC	SHEET NO: 40 / 40